

IS4

LESSON 24

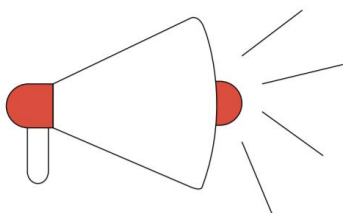
Artificial Intelligence of Things (AIoT)
IoT security
HW/SW environments for IoT/AIoT



Outline



- AIoT architecture and deployment choices
- IoT security and OTA updates
- Edge gateways, embedded runtimes, and TinyMLCloud
- ML services
- Intelligent vision systems for industry and logistics
- Industry 4.0 and 5.0 design concepts



Your opinion really matters!

By filling in the survey:

- you contribute **to the improvement** of your courses
- **your opinion** directly reaches the academic bodies in charge of **teaching quality**

Professors
who are motivated
to improve the
quality of their
courses



**The Joint
Committee** of students
and professors
which analyzes survey
results, identifies
problems and suggests
solutions.



the Education Board
where students and
professors evaluate
and publish the results,
propose and put into
practice concrete
measures to improve
the quality of courses.

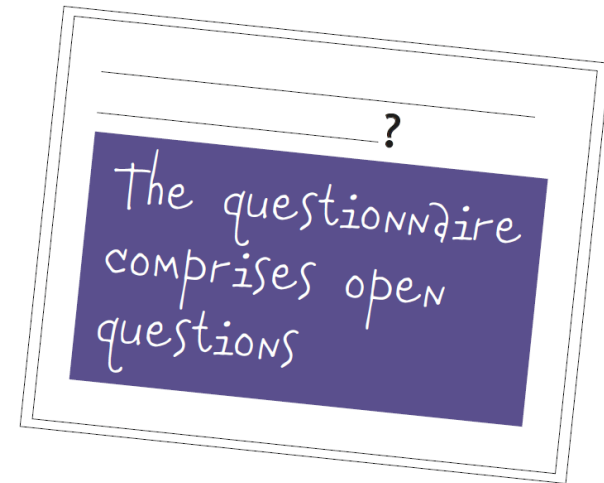


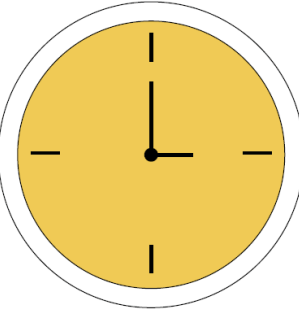
How to fill in the survey

Answer the questions carefully and clearly, be sure to point out to your professors and students' representatives problems and suggestions.

Remember that questionnaires are:

- ✓ completely anonymous
- ✓ must be filled in online for each of your courses





When to fill in the questionnaire

It's better to fill in the questionnaire after you have attended more than half of the classes in your course.

It's mandatory to fill it in before registering for the final exam in your course. **You can't register for the exam until you fill it in.**

Anyway, it's
better not to wait
until the last
minute!

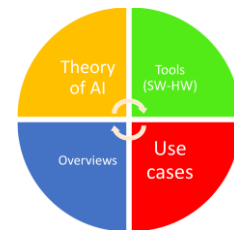




THEORY

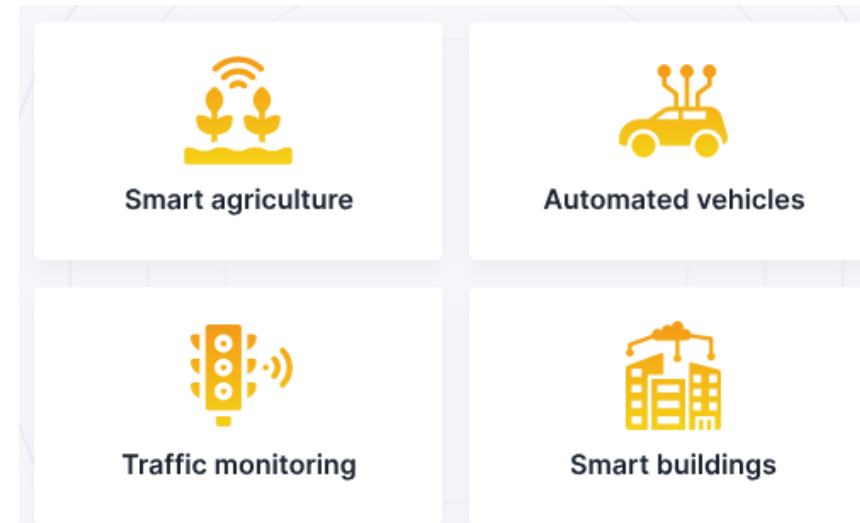
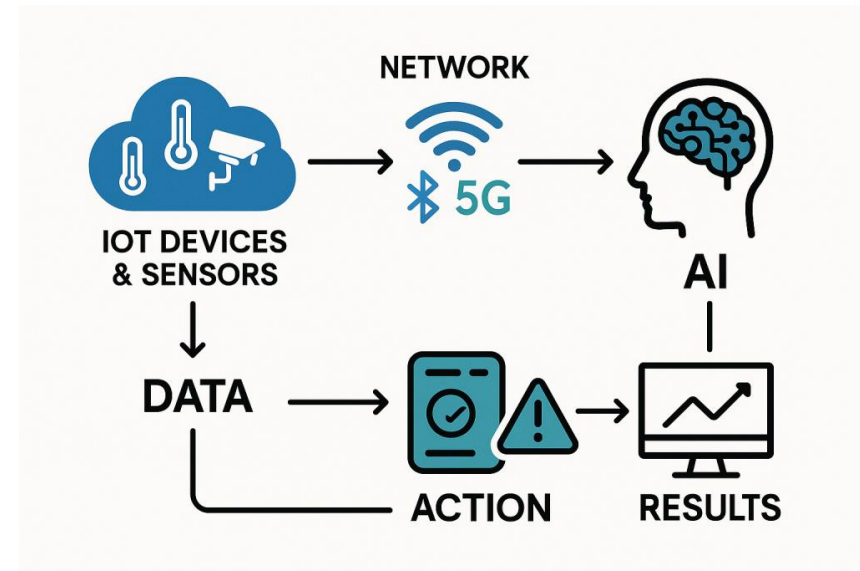
Artificial Intelligence Of Things (AIoT)

A new frontier for applications



Artificial Intelligence Of Things (AIoT)

- AIoT combines IoT infrastructure with AI models. IoT devices collect data from physical processes
- AI transforms these data into predictions, anomaly alerts, optimized actions, and decision support.
- AIoT technology create data "learning machines".
- Real time data is a key value of all AIoT use cases and solutions.
- In modern systems, part of the intelligence can run in the cloud, at the edge, or directly on the device, depending on latency, bandwidth, privacy, energy, and reliability constraints.



AIoT: HW

What is enabling the AI?

AIoT is enabled by devices and infrastructure able to sense, compute, communicate, and act.

- **Sensors**
cameras, IMUs, microphones, temperature, pressure, vibration, tyre sensors
- **Microcontrollers and embedded CPUs:**
ESP32, STM32, ARM Cortex-M, Raspberry Pi class devices
- **AI accelerators**
GPU, TPU, NPU, VPU, Edge TPU, Hailo, Jetson, FPGA
- **Connectivity**
Wi-Fi, Bluetooth, Zigbee, LoRaWAN, 4G/5G, Ethernet, fieldbus
- **Edge gateways**
industrial PCs or local machines running edge runtimes

AIoT: SW

What is enabling the AI?

Software enables interoperability, model deployment, monitoring, and control.

- **Embedded runtimes**
FreeRTOS, Eclipse ThreadX, embedded Linux
- **AI runtimes**
TensorFlow Lite, ONNX Runtime, vendor SDKs
- **Communication protocols**
MQTT, HTTP/REST, OPC UA, Modbus, CAN bus
- **Cloud/edge services**
AWS IoT Greengrass, Azure IoT, Google Cloud IoT/Vertex AI style platforms
- **APIs and MLOps tools**
model endpoints, OTA updates, logging, monitoring, retraining pipelines

AIoT improvements

While developing your AI+IoT application you should keep in mind in your design to create a system that can be

- more **robust**
 - Self calibration, self diagnosis
 - Better noise reduction due to adaptive capabilities
- more **secure**
 - Information are provided with accuracy and assessing quality of the data
 - Temperature = 152 degree (sensor error, or engine is melting?)
- more **scalable**
 - Supporting different protocols and data format can talk and work with many different applications/devices/services

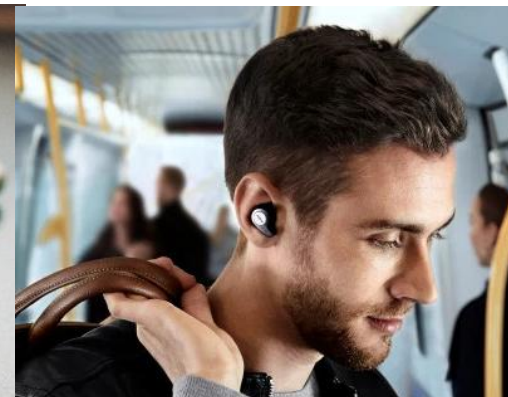
In brief → more **valuable**

- Same components, but the **SW is making the device more attractive**

Example of AIoT: Smart speakers

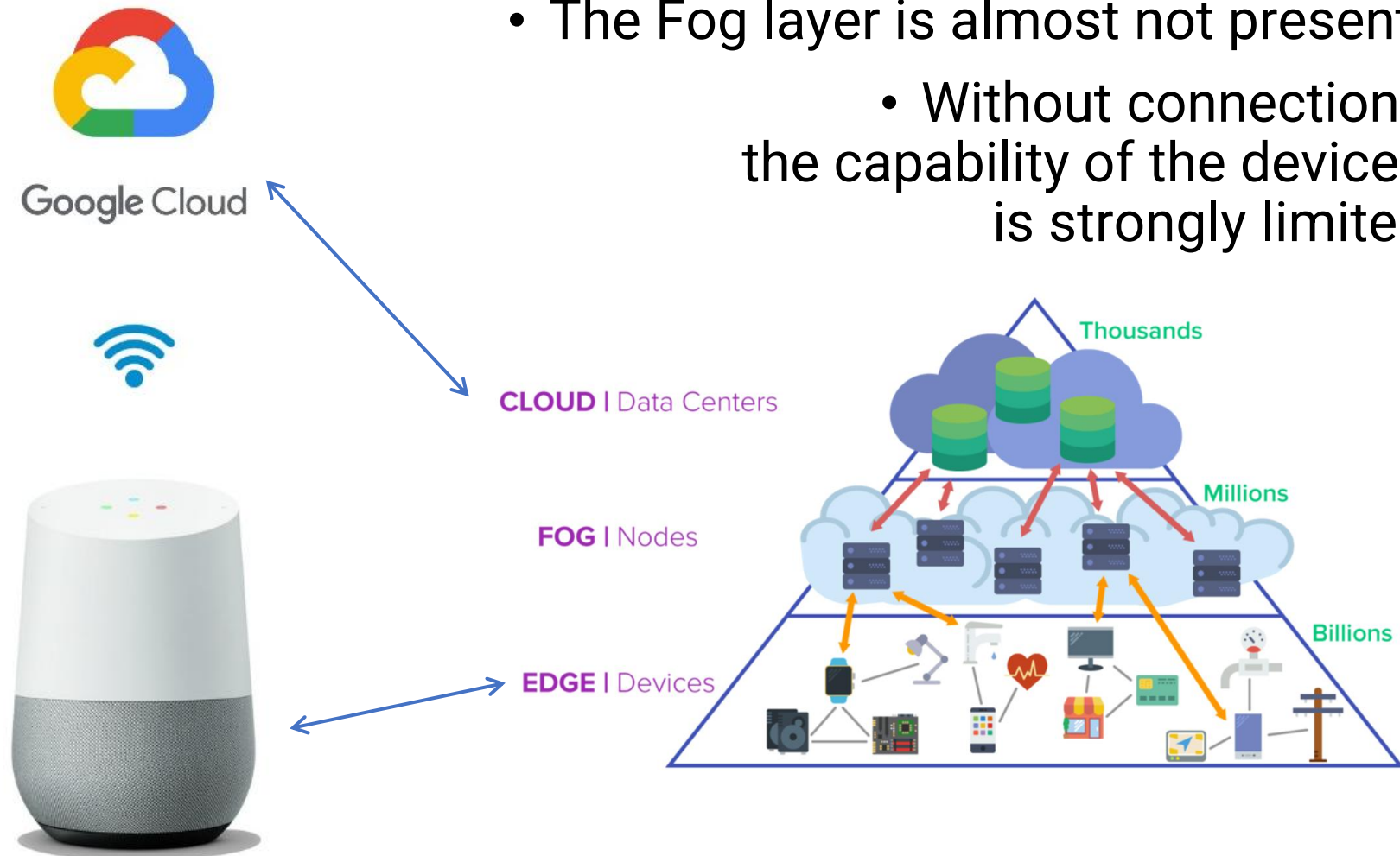


- AI enabled smart speaker can process the trigger word using a Natural Language Processing (NLP) model locally rather than sending all the voice capture to the cloud for processing.
- Type: Dynamic Data Source
- Hybrid approaches
 - Local processing → activation,
 - Remote processing → NLP (now also locally processed)



Example of AIoT: Smart speakers (2)

- The Fog layer is almost not present.
 - Without connections the capability of the devices is strongly limited



Example of AIoT: Intelligent & connected **lighting**

Edge Devices



 **zigbee**

(Bridge)



Trad.
IoT

AIoT

Edge Device



Bluetooth
(no bridge)

Edge Device

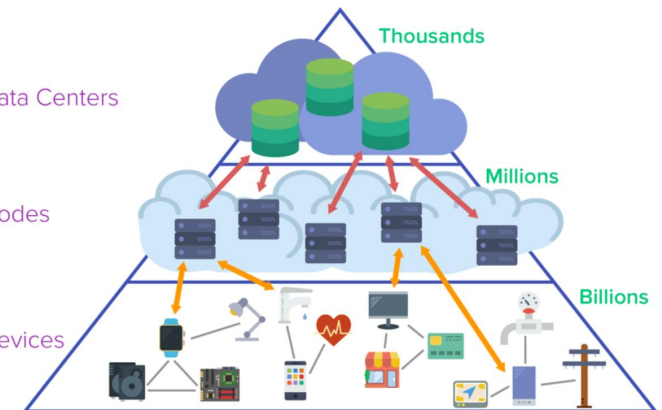


Source of commands (NLP via cloud)
and directly to the bridge

CLOUD | Data Centers

FOG | Nodes

EDGE | Devices

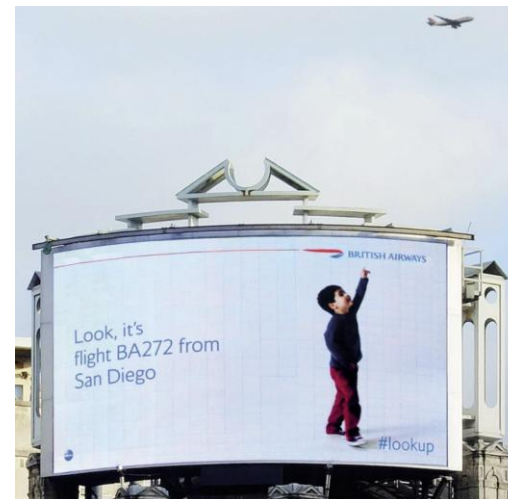


Examples of AIoT for Smart Retail/Ads Boards

AIoT typically use computer vision capabilities and sensor to gather intel about customers, sensors, vehicles, etc.



The light bulb remains dark until the moment a passerby walks beneath it. Then the bulb, thanks to motion sensors, lights up. It appears to people looking at the board as though the person walking under just had a fantastic idea — the light bulb moment



British Airways
Piccadilly Circus
#LookUp campaign



THEORY

Security of IoT

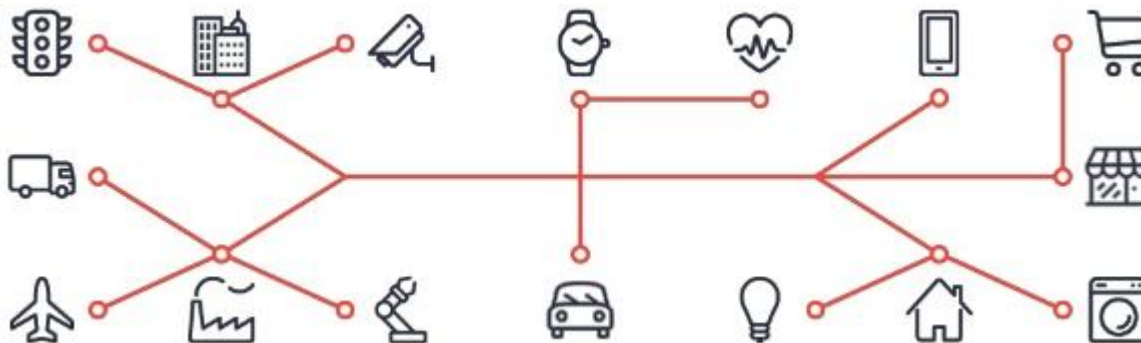
A new frontier for applications



Security of IoT

- The number and the devices and their relevance poses huge security problems
 - Just think to
 - Heart pacemakers
 - Traffic light systems
 - Car control-driving systems

Complex interactions →
More security issues

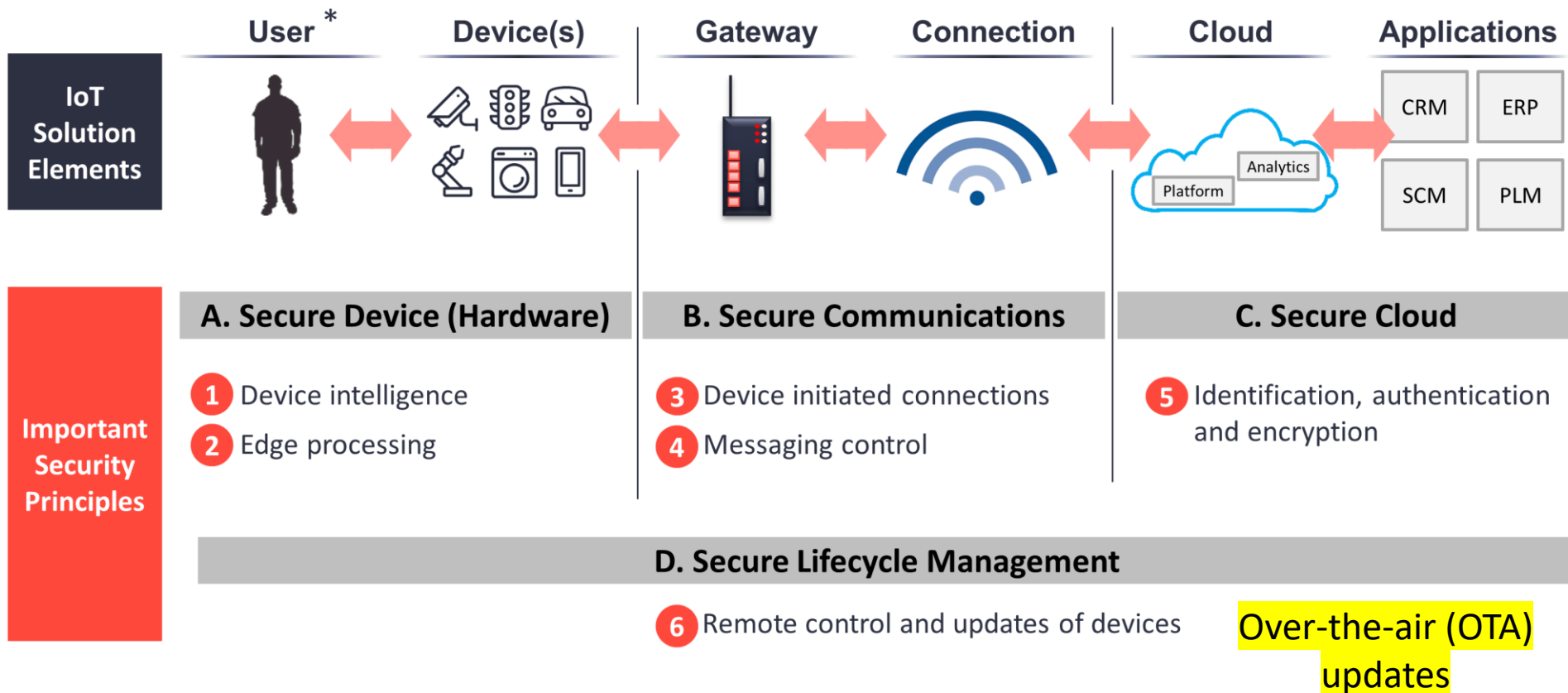


Data Sources

- Passive data
- Active data
- Dynamic data

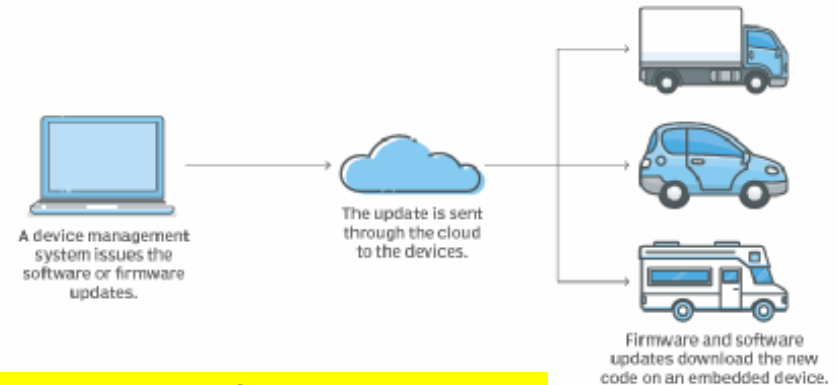


Six principles of IoT Cyber security in the stack



OTA

Over-the-air update process for IoT devices



- OTA (Over-the-Air) is a standardized method for delivering software and firmware updates to IoT devices remotely, without the need for physical access.
- This technology has become a cornerstone of modern IoT ecosystems, enabling seamless enhancements and security patches for a wide range of devices, such as connected cars, smart sensors, and industrial equipment.
- Standard for IoT updates (e.g., cars, sensors)
- To ensure data integrity and confidentiality during transmission, leverage secure communication protocols like
 - HTTPS (Hypertext Transfer Protocol Secure)
 - MQTT (Message Queuing Telemetry Transport),
 - lightweight messaging protocol specifically designed for efficient communication in resource-constrained environments)
 - TLS (Transport Layer Security)
 - cryptographic protocol designed to secure communications over networks by encrypting data, authenticating endpoints, and ensuring integrity

IoT security recent standards

Cyber Resilience Act (EU, 2025)

Mandatory security:

- EU regulation enforcing cybersecurity for all digital products (hardware/software) sold in the EU market.
- CE marking to certify compliance.
- Lifecycle support (updates, vulnerability reporting).

IoT security recent standards

Cyber Resilience Act (EU, 2025)

- **Mandatory "security by design"** for manufacturers.
 - **Risk Assessment and Mitigation**
 - Cybersecurity risk assessment from the initial design step
 - **Secure Default Settings**
 - Products must ship with secure configurations "out of the box." (e.g., strong password as default)
 - **Protection Against Common Threats**
 - (e.g., TLS for data transmission) and safeguarding against exploits like injection attacks.
 - **Minimized Attack Surface**
 - Limit unnecessary features, open ports, or exposed interfaces that could be exploited.
 - **Secure Development Practices**
 - (e.g., avoiding hard-coded credentials).
 - **Lifecycle Security**
 - software/firmware updates to address vulnerabilities.
 - **Transparency and Documentation**

IoT security recent standards

NIST (USA) IoT Cybersecurity Framework

(2024)

- **Voluntary** U.S. framework by NIST to secure IoT devices and ecosystems.
- **Identify, Protect, Detect, Respond, Recover**

Example: a smart factory with IoT sensors

- **Identify:** Map all sensors and their data outputs.
- **Protect:** Encrypt sensor communications.
- **Detect:** Monitor for unusual sensor activity.
- **Respond:** Isolate a hacked sensor.
- **Recover:** Update firmware and reinforce security.



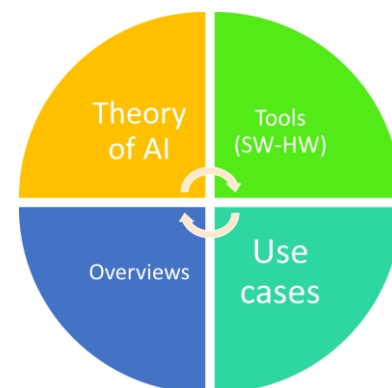


Toolboxes

Amazon

AWS for IOT

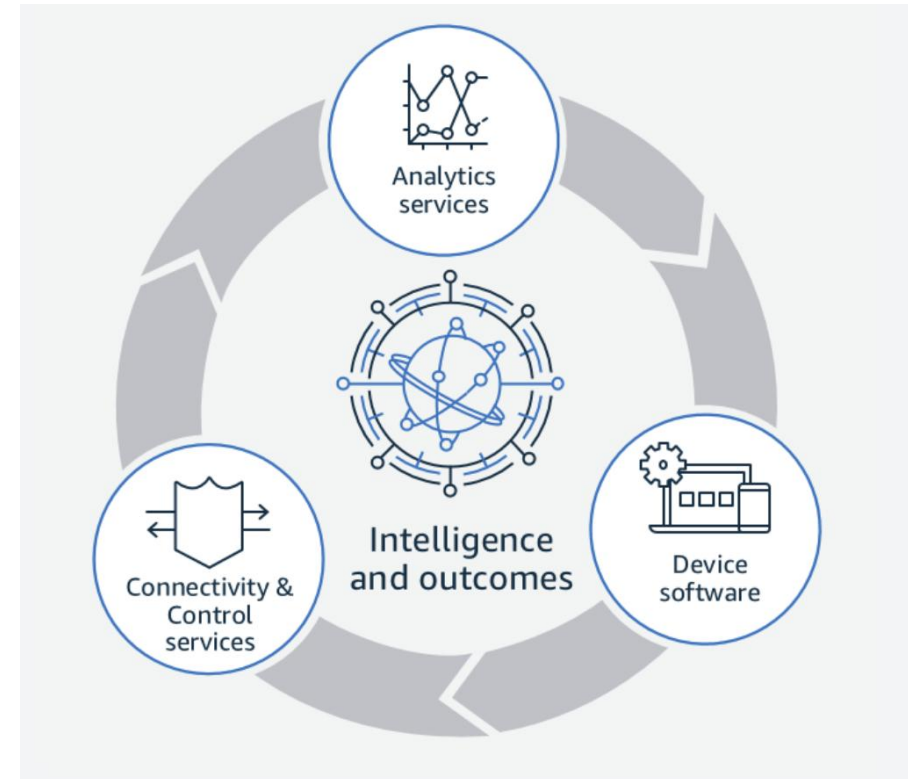
A platform at your disposal



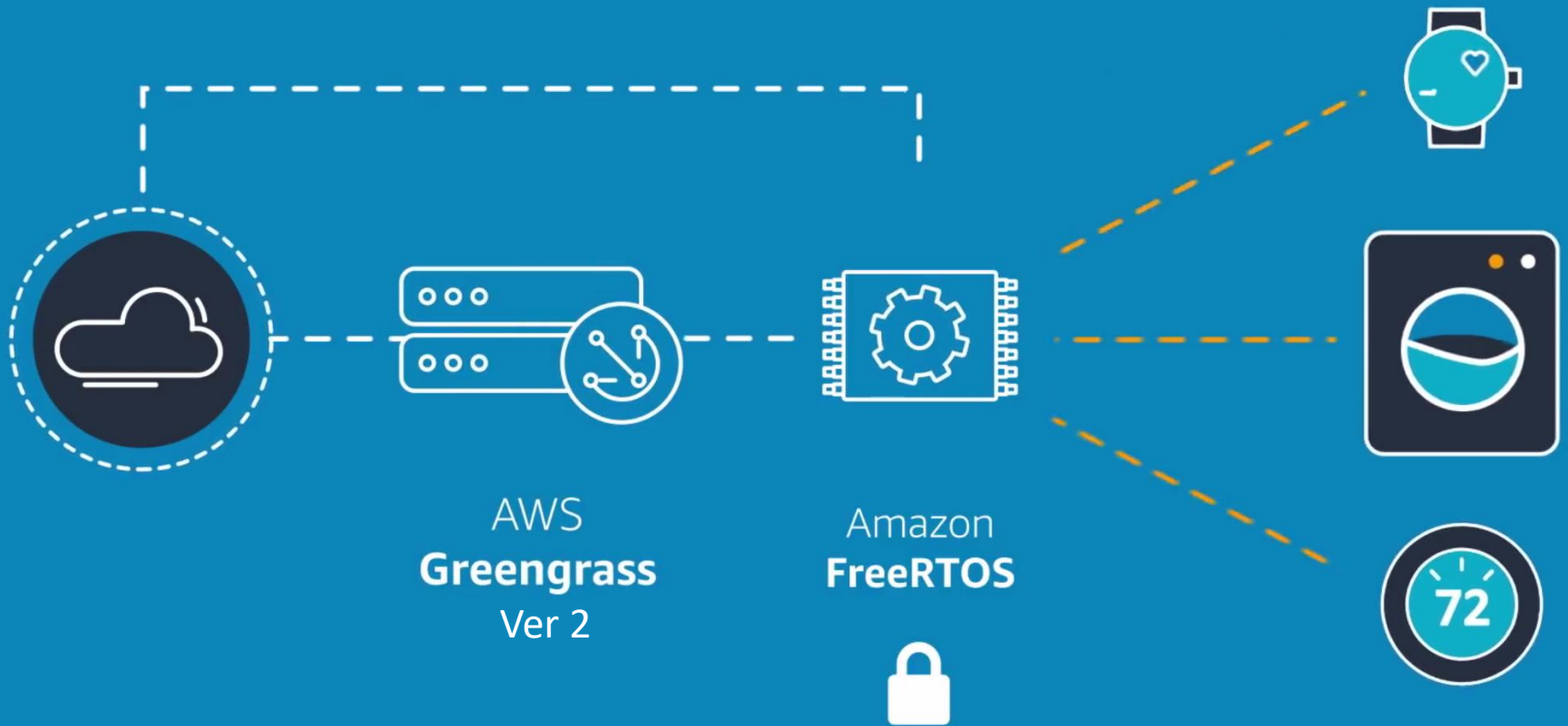
Amazon AWS IoT



- IoT services for industrial, consumer, and commercial solutions
- Broad and deep
 - From the edge to the cloud, **device software** (FreeRTOS) provides local **data collection and analysis** (AWS IoT Greengrass)
 - Data management + Rich analytics

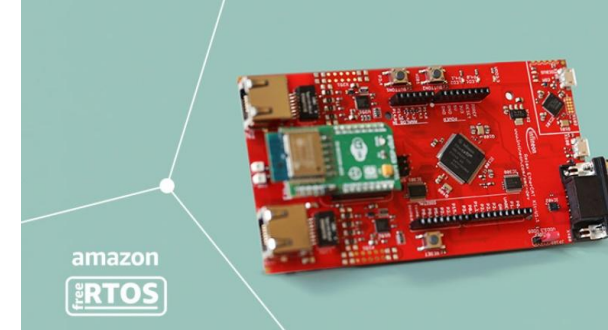


The basic idea



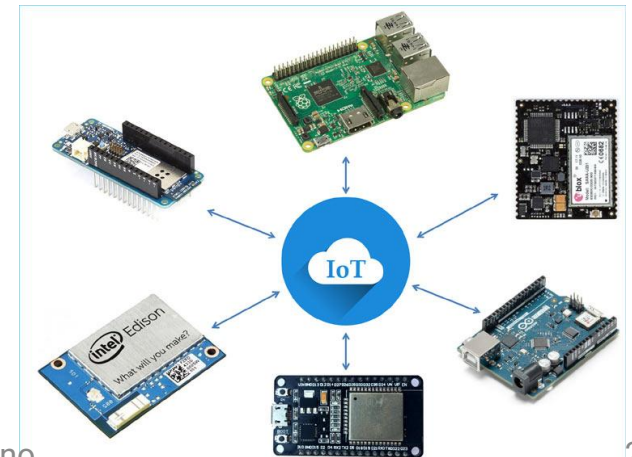
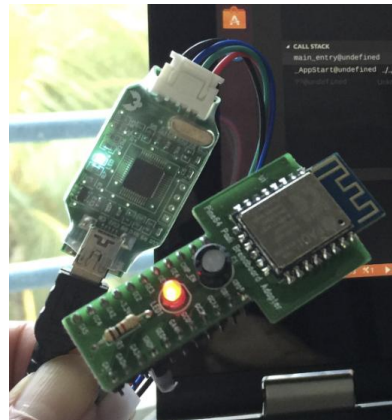
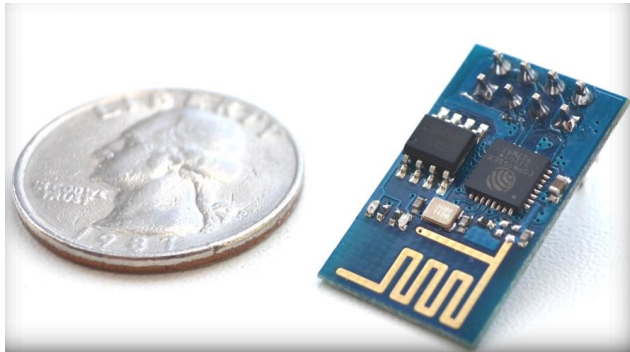
FreeRTOS for embedded AIoT devices

Real-time operating system for microcontrollers

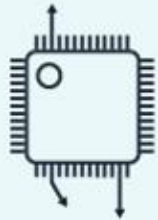


Or a Raspberry Pi 3B.

- **FreeRTOS** is an open source, real-time operating system for microcontrollers that makes small, low-power edge devices easy to program, deploy, secure, connect, and manage.
- A **microcontroller** contains a simple, resource-constrained processor that can be found in many devices, including appliances, sensors, fitness trackers, industrial automation, and automobiles.



Design steps



Choose a supported microcontroller

Select a supported microcontroller from Amazon FreeRTOS Qualification Program



Download OS and libraries

Download the FreeRTOS kernel and libraries for security, cloud and local connectivity, and updateability through the Amazon FreeRTOS console



Develop app

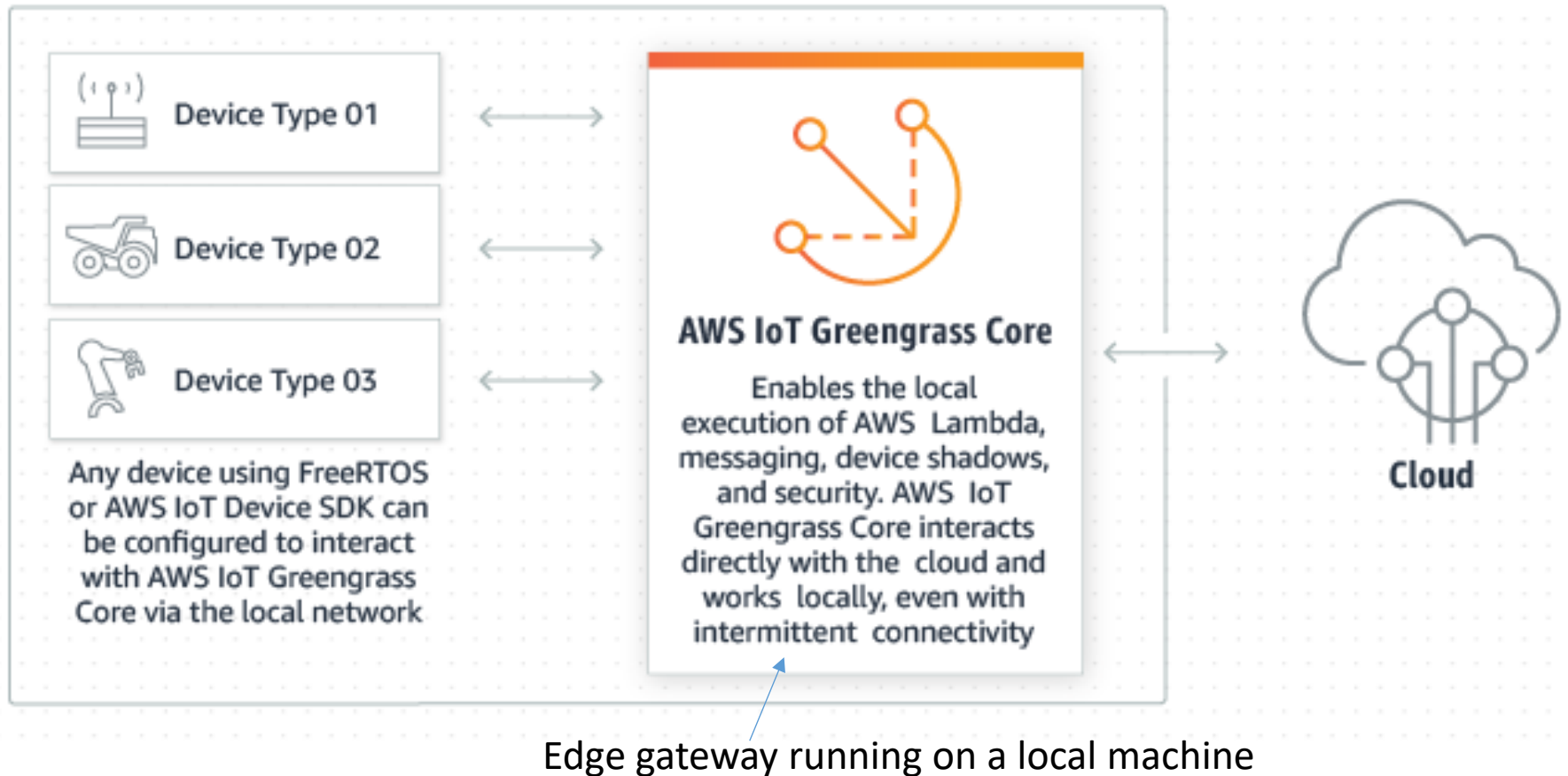
Develop IoT applications for microcontroller-based devices with convenient APIs



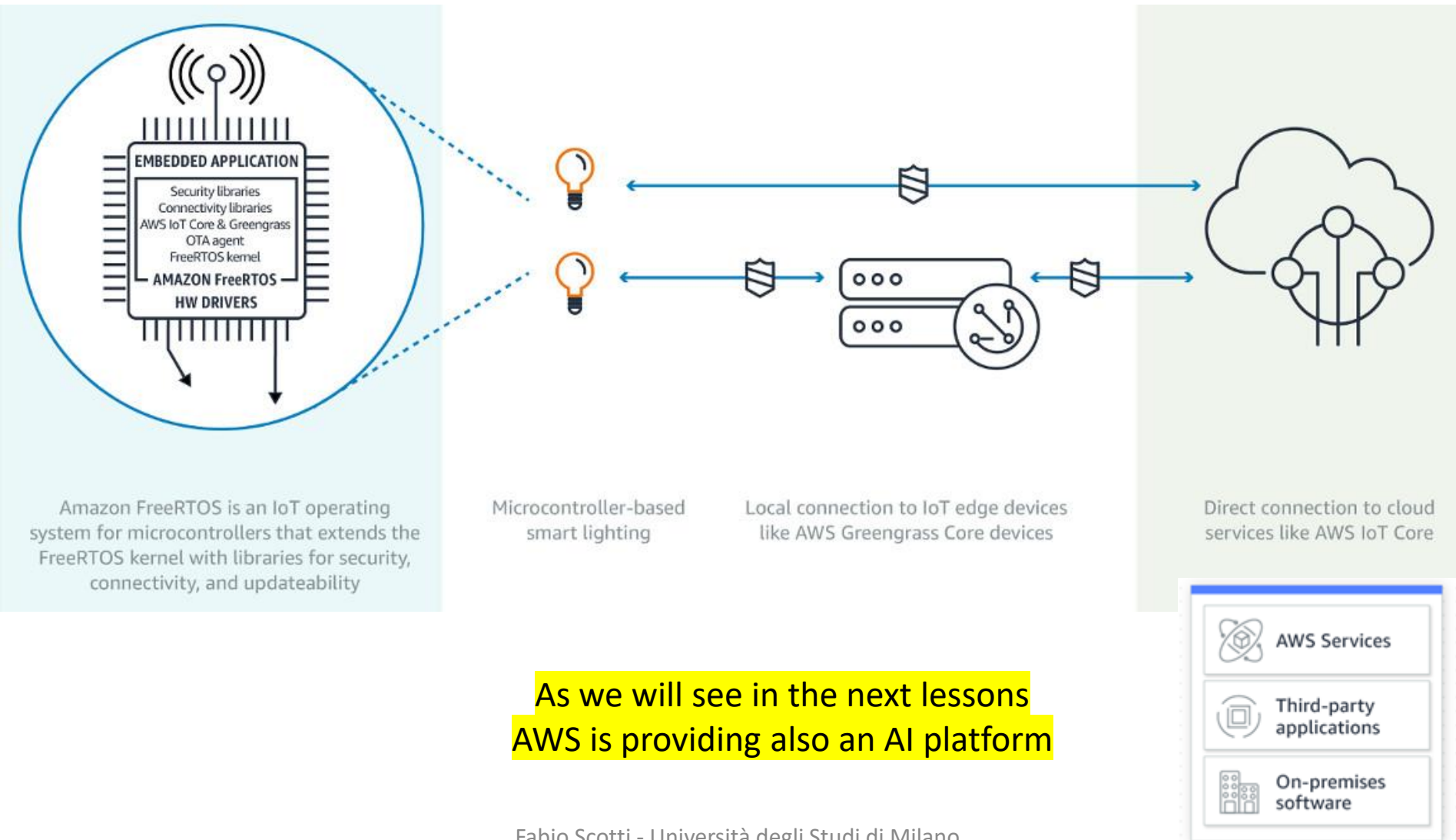
Deploy connected device

Deploy and maintain connected microcontroller-based devices at scale. Easily connect to local gateways like AWS Greengrass Core devices or to AWS cloud services like AWS IoT Core

AWS IoT Greengrass



From a «lamp» to 3rd party app



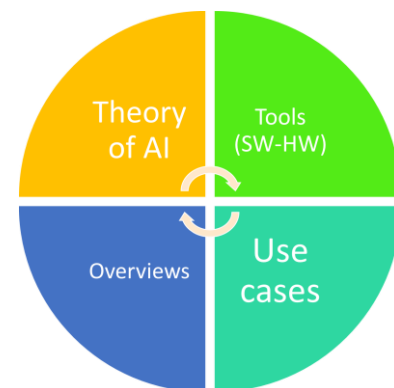


Toolboxes

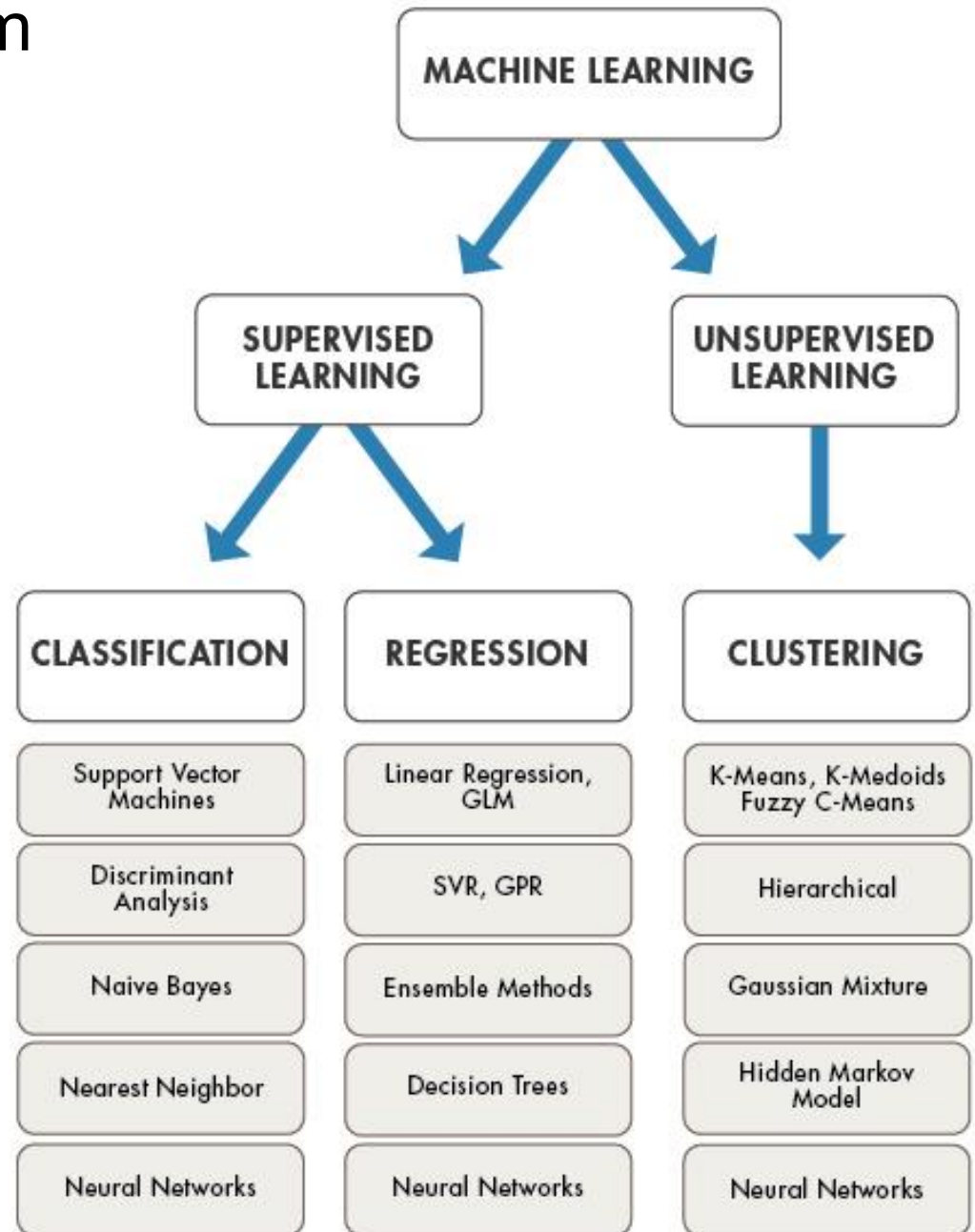
Matlab & IOT

Focusing on embedded boards

Arduino

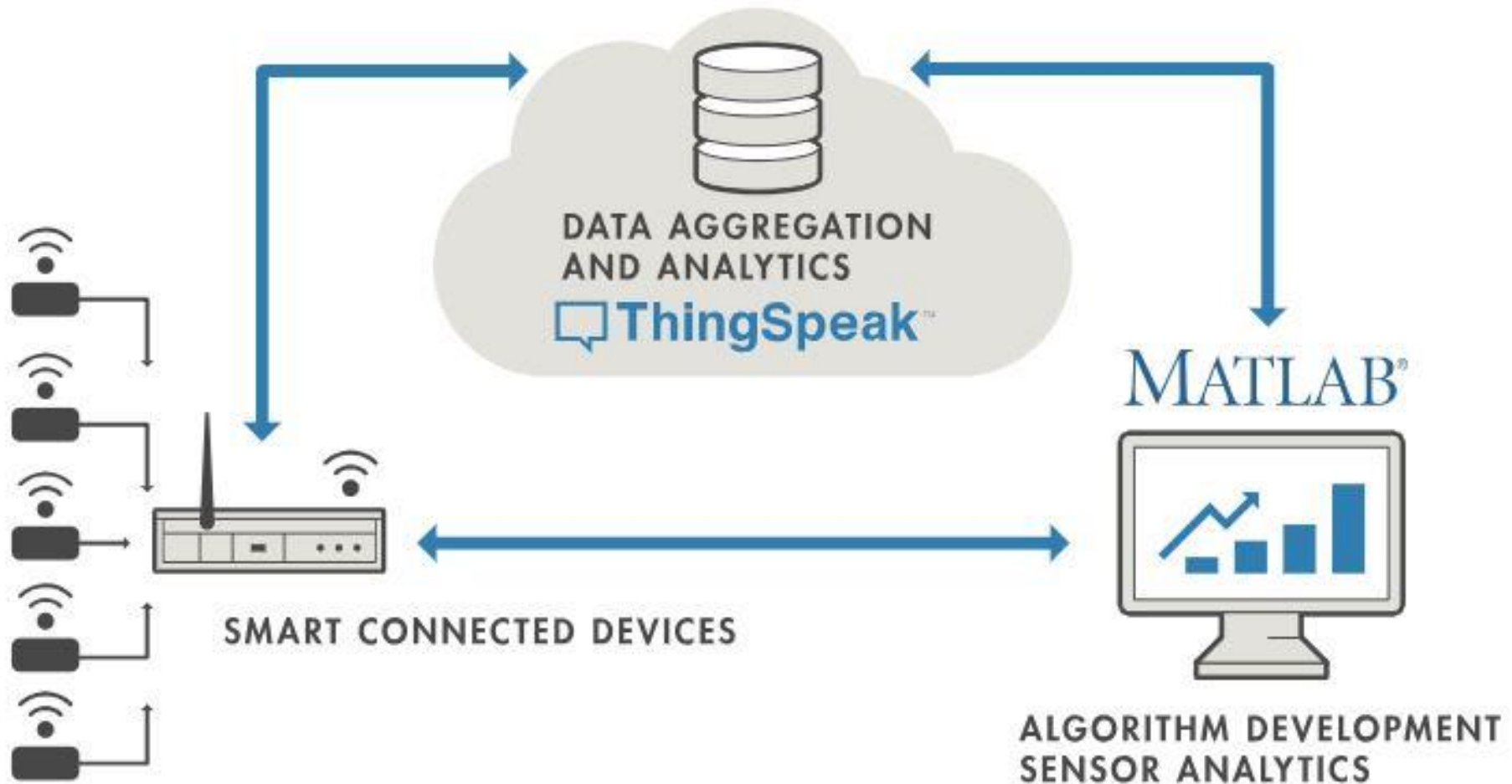


A very good ML platform
But not only!



and many others!

Matlab ThingSpeak



Off-line/On-line functions

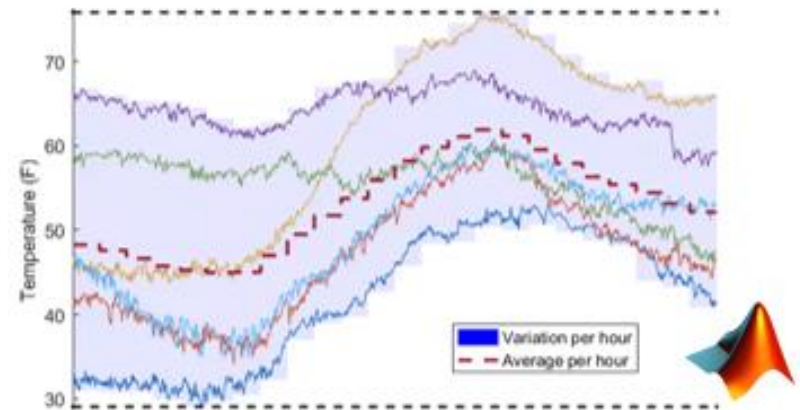
1 While offline, collect sensor data



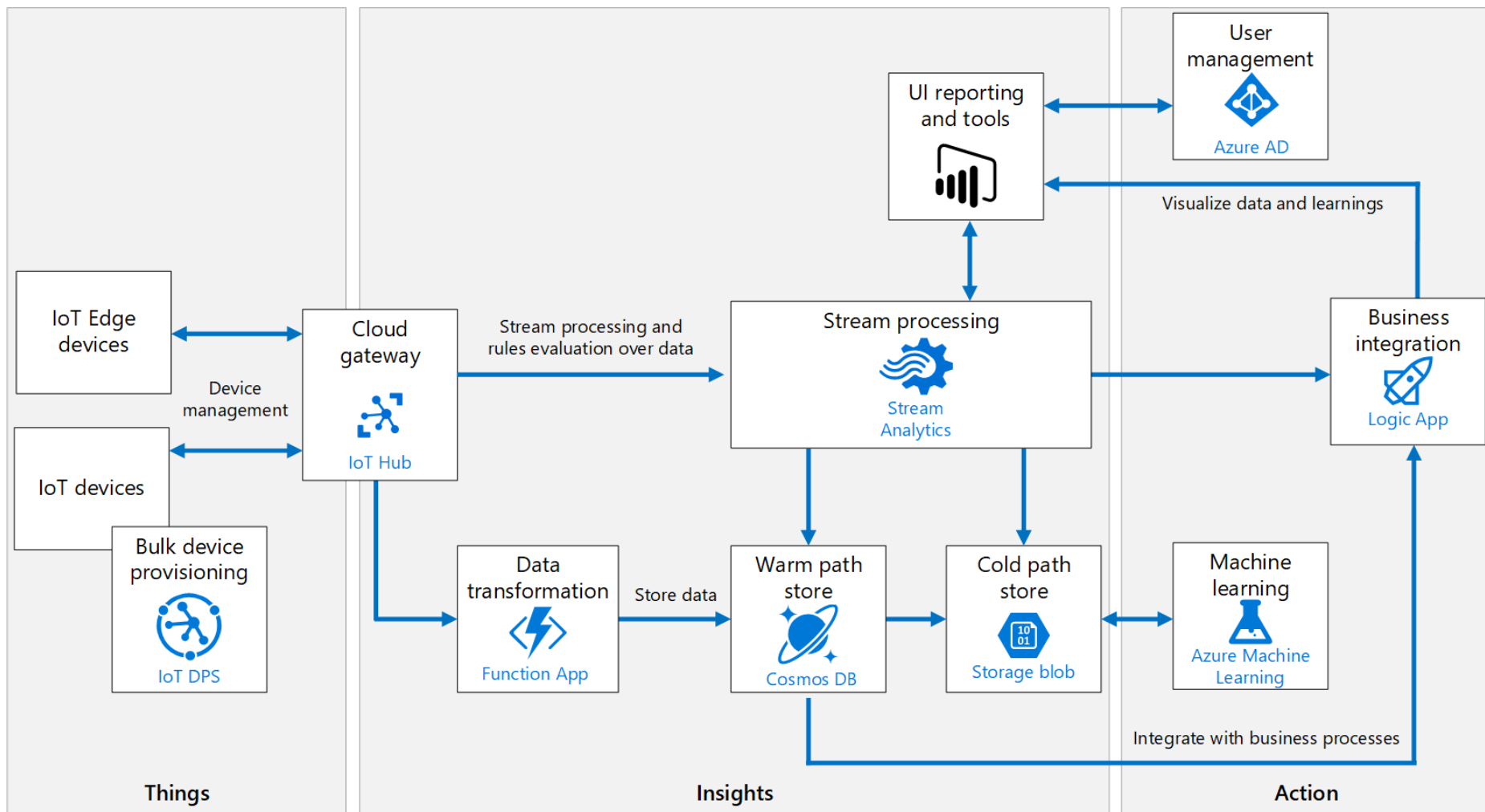
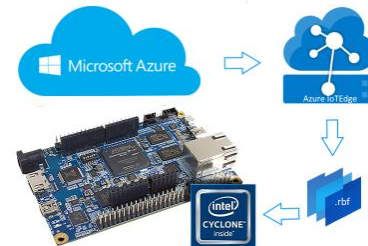
2 Periodically connect to network and send collected data to ThingSpeak.com all at once using **bulk_update**



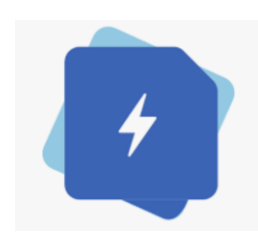
3 Analyze data using **thingSpeakRead** in MATLAB



Microsoft Azure IoT



Eclipse ThreadX, formerly Azure RTOS



- Embedded development suite including a small but powerful operating system that provides reliable, ultra-fast performance for resource-constrained devices.
- Easy-to-use
- Portable: ThreadX supports the most popular 32-bit microcontrollers and embedded development tools (reuse of previous projects and knowledge).
- Market-proven: more than 6.2 billion devices worldwide.
- Safe: Ipsec (IP level), TLS and DTLS (socket level)
- The firmware can be updated by **over-the-air** using automatic device management



Google Coral

Build beneficial and privacy preserving AI

- A platform developed by Google for building and deploying artificial intelligence (AI) on the edge without relying on cloud processing.
- It consists of hardware components, software tools, and pre-trained machine learning (ML) models designed to enable fast and efficient AI processing locally on embedded systems.



Efficient

Balance power and performance with local, embedded applications.



Fast

Run lightning-fast AI at industry-leading inference speeds for embedded devices.



Private

Keep user data private by performing all inferences locally. You decide when data is stored or transferred.



Offline

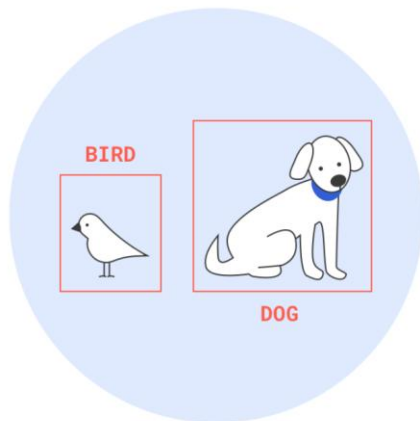
Deploy in the field where connectivity is limited.

The image displays three different Coral Dev Board configurations. On the left is the 'Dev Board', a single-board computer with a large black cooling fan and various ports. In the center is the 'USB Accelerator', a small white rectangular device with a USB Type-C connector. On the right is the 'Dev Board Mini', a smaller single-board computer without a fan.

-
- A collection of various Coral Dev Boards, including the Coral Dev Board, Coral Dev Board Mini, Coral Dev Board Micro, Coral Dev Board Edge, Coral Dev Board Pro, and Coral Dev Board Nano. The boards are shown in different orientations and sizes, highlighting their diverse form factors and components like fans, heat sinks, and connectors.

Google Coral

- With a vaste range of pretrained models to embed in the HW.
- **Very fast inference** (Limited on-device transfer learning is possible only in specific cases.).



Object detection

Draw a square around the location of various recognized objects in an image.



Pose estimation

Estimate the poses of people in an image by identifying various body joints.

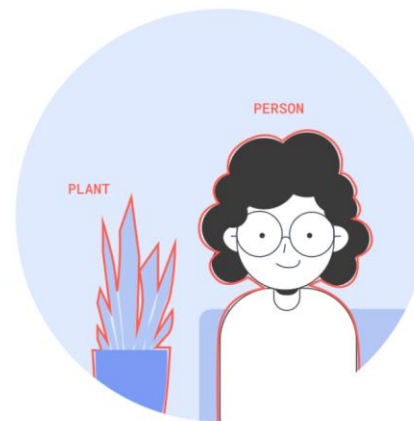


Image segmentation

Identify various objects in an image and their location on a pixel-by-pixel basis.



Key phrase detection

Listen to audio samples and quickly recognize known words and phrases.



Toolboxes

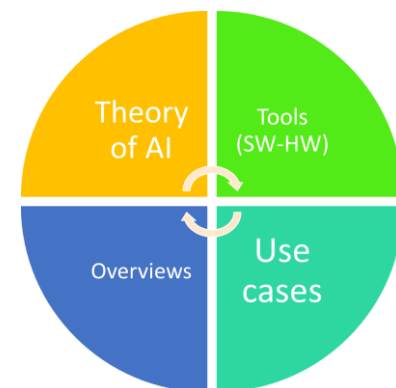
Boards for IOT



Focusing on embedded boards

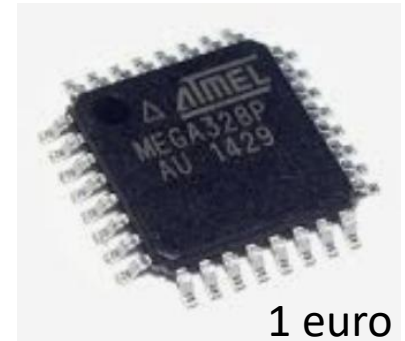
Arduino

Industrial boards and accelerators



Some IoT devices

- From a very small microcontroller ATmega328P (Arduino UNO)
8bit 32 KB flash



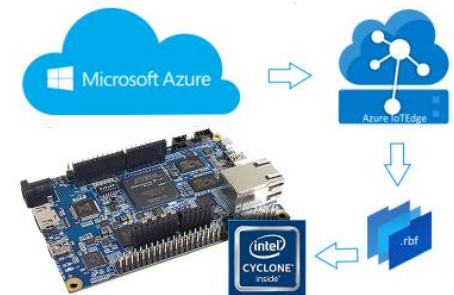
- Matlab yes; AWS IoT no

- Atmel ATSAMW25 SoC designed for IoT projects and devices (**MKR1000** Arduino)



- ARM larger processor 32bit, 64bit
 - Matlab yes; AWS IoT yes

- FPGA + Huge microcontroller + Vision cameras



Boards and AIoT



Raspberry Pi 4 B

4x Cortex-A72 CPU
VideoCore VI GPU
32 GFLOPS
1.5 GHz - 1/2/4/8 GB
★★★★☆
€ 40/€ 40/€ 60/€ 85

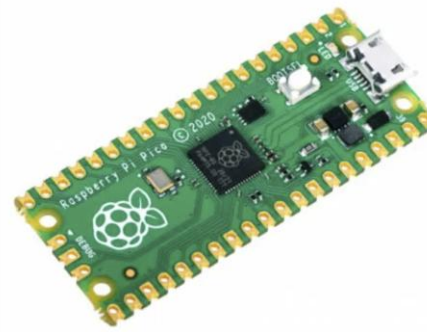
The successor to the Raspberry Pi 3 with a slightly faster processor, USB 3.0 and GigaEhternet.



Intel Neural Stick 2

Intel Movidius Myriad X
16 SHAVE cores
1 TOPS
☆☆☆☆☆
★☆☆☆☆
€ 87

Special Intel neural network USB 3 dongle for PC and single boards like Raspberry Pi. Accelerates tensor arithmetic enormously. Fully supported by OpenCV.



Raspberry Pi Pico

2x Cortex-M0+ CPU
-
133 MHz - 264 KB
☆☆☆☆☆
€ 5

Just some I/O, an RP2040 MCU and 2 MB of flash. Can it be used for deep learning? Barely. However, TensorFlow TinyML has some examples [here](#).

Boards and AIoT (2)



Jetson Nano 2 GByte

4x Cortex-A57 CPU

128x CUDA

1.88 TOPS

1.43 GHz - 2 GB

★★★★☆

★★★★☆

€ 50

Stripped Nano version to compete with the Raspberry. One USB 3 port, a single camera and 2 GByte of RAM. A good start for your first deep learning steps, as long as your model fits the 2 GB RAM. [More NVIDIA boards.](#)



Google SOM

4x Cortex-A53 + 1x Cortex M4 CPU

GC7000 Lite 3D GPU

4.0 TOPS NPU

1.5 + 1.0 GHz - 1 GB

★★★★☆

★★★★☆

€ 105

Single tiny (40x48 mm) pluggable module with full I/O and the [Edge TPU](#) accelerator.

Raspberry Pi AI Kit



- The Raspberry Pi AI Kit bundles the Raspberry Pi M.2 HAT+ with a Hailo AI acceleration module for use with Raspberry Pi 5. It provides an accessible, cost-effective, and power-efficient way to integrate high-performance AI.
- The AI module is a **13 tera-operations per second (TOPS)** neural network inference accelerator built around the Hailo-8L chip
 - 8-bit integer (INT8) precision

Boards and AIoT

TOPS= Tera Operations Per Second.

 We will use/retrain them in the course

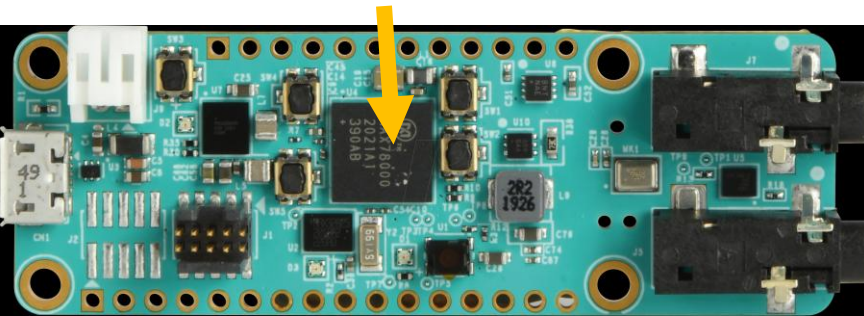
Model	Framework	Raspberry Pi (TF-Lite)	Raspberry Pi (ncnn)	Raspberry Pi Intel Neural Stick 2	Raspberry Pi Google Coral USB	JeVois	Jetson Nano	Google Coral
EfficientNet-B0 (224x224)	TensorFlow	14.6 FPS (Pi 3) 25.8 FPS (Pi 4)	-	95 FPS (Pi 3) 180 FPS (Pi 4)	105 FPS (Pi 3) 200 FPS (Pi 4)	-	216 FPS	200 FPS
 ResNet-50 (244x244)	TensorFlow	2.4 FPS (Pi 3) 4.3 FPS (Pi 4)	1.7 FPS (Pi 3) 3 FPS (Pi 4)	16 FPS (Pi 3) 60 FPS (Pi 4)	10 FPS (Pi 3) 18.8 FPS (Pi 4)	-	36 FPS	18.8 FPS
MobileNet-v2 (300x300)	TensorFlow	8.5 FPS (Pi 3) 15.3 FPS (Pi 4)	8 FPS (Pi 3) 8.9 FPS (Pi 4)	30 FPS (Pi 3)	46 FPS (Pi 3)	30 FPS	64 FPS	130 FPS
SSD Mobilenet-V2 (300-300)	TensorFlow	7.3 FPS (Pi 3) 13 FPS (Pi 4)	3.7 FPS (Pi 3) 5.8 FPS (Pi 4)	11 FPS (Pi 3) 41 FPS (Pi 4)	17 FPS (Pi 3) 55 FPS (Pi 4)	-	39 FPS	48 FPS
Binary model (300x300)	XNOR	6.8 FPS (Pi 3) 12.5 FPS (Pi 4)	-	-	-	-	-	-
 Inception V4 (299x299)	PyTorch	-	-	-	3 FPS (Pi 3)	-	11 FPS	9 FPS
Tiny YOLO V3 (416x416)	Darknet	0.5 FPS (Pi 3) 1 FPS (Pi 4)	1.1 FPS (Pi 3) 1.9 FPS (Pi 4)	-	-	2.2 FPS	25 FPS	-
 OpenPose (256x256)	Caffe	4.3 FPS (Pi 3) 10.3 FPS (Pi 4)	-	5 FPS (Pi 3)	-	-	14 FPS	-

Industrial boards for AIoT

Many other companies are now producing industrial boards with Deep Learning model accelerators such as ST with **STM32CubeMX**

- Automatic conversion of **pre-trained Neural Network**
- Integration of generated optimized library into the user's project
- Azure compliant

Example: MAX78000



AIoT in Logistics and general apps



- Intelligence on the sensor
- Intelligent asset tracking with machine learning
 - You can build decision tree for the sensor without writing a single line of code
- LSM6DSOX 6-axis inertial module with machine learning core and explain how you can build intelligent asset tracking logic into this sensor with simple steps that will have you developing quickly and getting to market ahead of time.

Intro to LSM6DSOX:
Machine Learning Core



Matlab white paper for IoT

WHITE PAPER

Using MATLAB with Big Data
from Sensors and IoT Devices

A short paper (not in the exam) offering a very good overview and examples of projects and implementations



Matlab white paper for IoT

For example, how to create a Predictive Maintenance Alarm system. These concepts can be applied to many different applications a contexts.

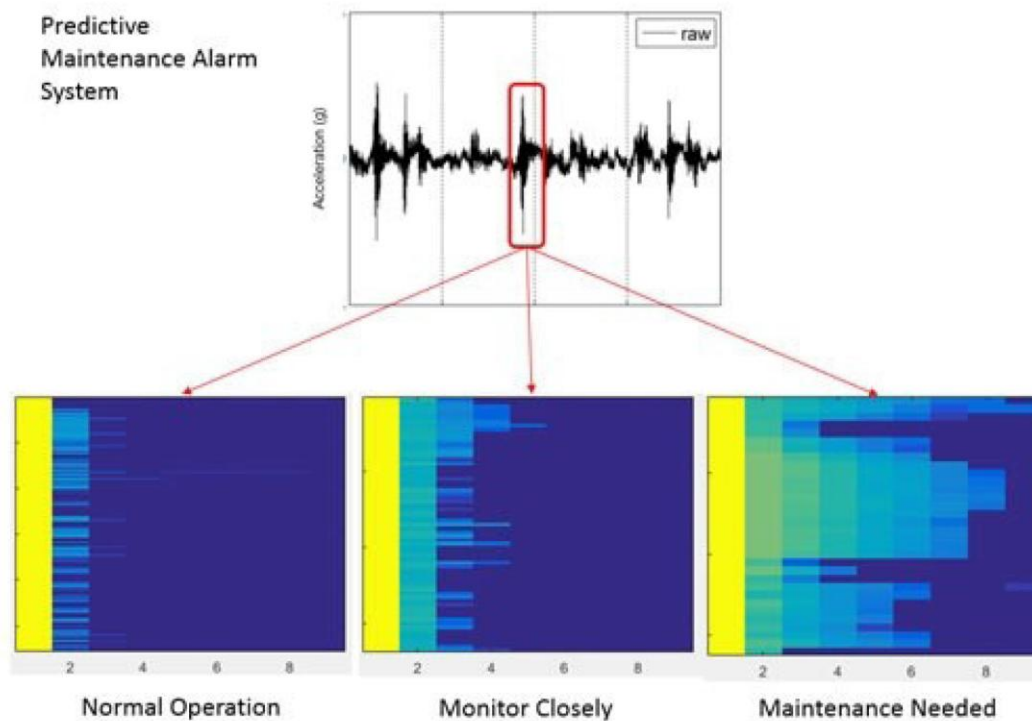


Figure 1. Baker Hughes' predictive maintenance alarm system, based on MATLAB.

Main points



- Usecase: The Pirelli Cyber Tyre IoT Architecture
- Security of IoT
- Artificial Intelligence Of Things (AIoT)
- Toolboxes:
 - Introduction to Matlab
 - Matlab for IoT and AIoT
 - The Arduino use case
 - Amazon AWS IoT, FreeRTOS
 - Azure RDT, and W11 IoT
- Industrial Board for AIoT

LESSON 24

Part B

Machine Learning Software and Cloud Services
Designing for Industry 4.0 and 5.0 applications

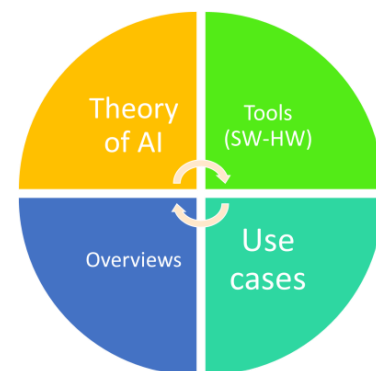




Toolboxes

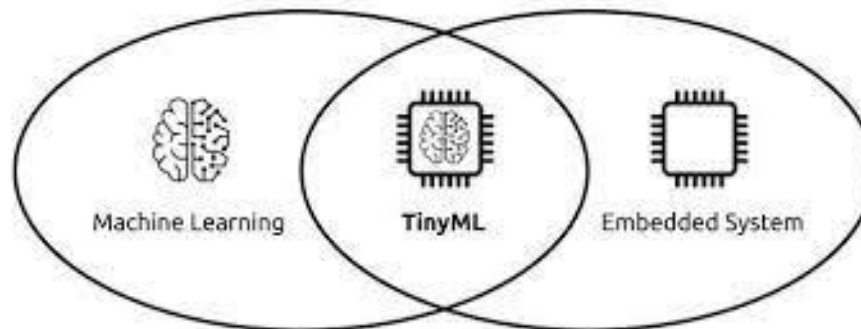
TinyML

Machine Learning for very small device



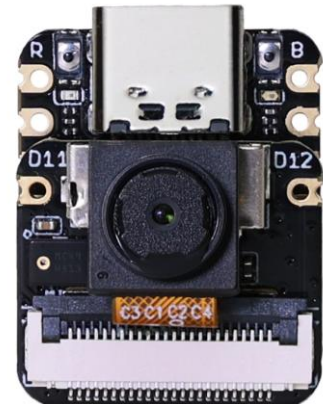
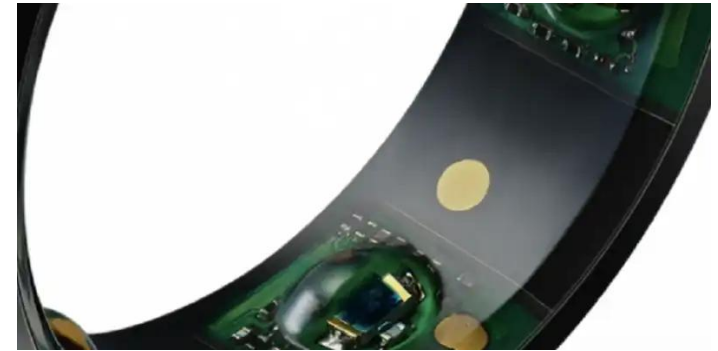
Tiny ML

- a subset of machine learning that enables machine learning models to run on small, low-power devices, such as microcontrollers.
- TinyML models are typically trained on powerful cloud servers and then deployed to edge devices, where they can perform inference with low latency and power consumption.



Tiny ML example

- **Smart thermostat** that can learn your heating and cooling preferences and adjust the temperature accordingly.
- **Wearable fitness tracker** that can track your activity levels and sleep quality.
- **Smart speaker** that can recognize your voice and respond to your commands.
- **Security camera** that can detect intruders and send you an alert.
- **Smart factory machine** that can monitor its own performance and predict when maintenance is needed.



Tiny ML applications

• Wide Applications

Smart Retail



Personalized Healthcare



Smart Home



Smart Manufacturing



Precision Agriculture



Autonomous Driving



- **Wearable devices**

TinyML can be used to develop wearable devices, such as smartwatches and fitness trackers, that can perform tasks such as activity tracking, heart rate monitoring, and gesture recognition.

- **Industrial IoT**

TinyML can be used to develop industrial IoT devices that can perform tasks such as predictive maintenance and anomaly detection.

- **Smart homes**

TinyML can be used to develop smart home devices that can perform tasks such as motion detection, voice recognition, and energy management.

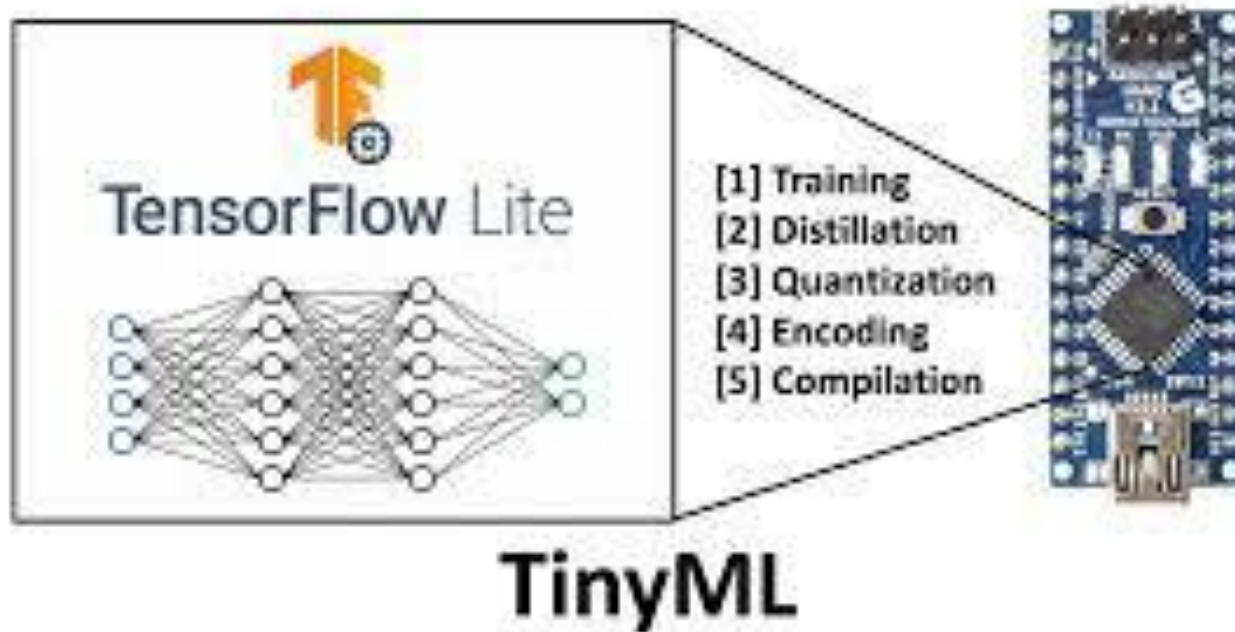
- **Other applications**

TinyML can also be used to develop applications in other areas such as agriculture, healthcare, and transportation.

Traditional VS TinyML

	Traditional	Tiny device
Computing 	1 GHz – 4 GHz	1 MHz – 400 MHz
Memory 	512 MB – 64 GB	2 KB – 512 KB
Storage 	64 GB – 4 TB	32 KB – 2 MB
Power 	30 W – 100 W	150 μ W – 23.5 mW

TensorFlow Lite for compact implementations



TensorFlow Lite is a lightweight version of TensorFlow designed for mobile and embedded devices. It enables **on-device machine learning (ML) inference with low latency and a small binary size**, making it ideal for resource-constrained environments like smartphones, IoT devices, and microcontrollers.

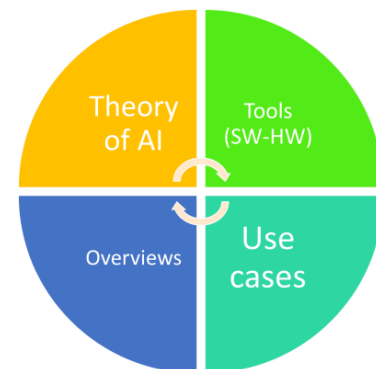


Toolboxes

Cloud Services

for ML

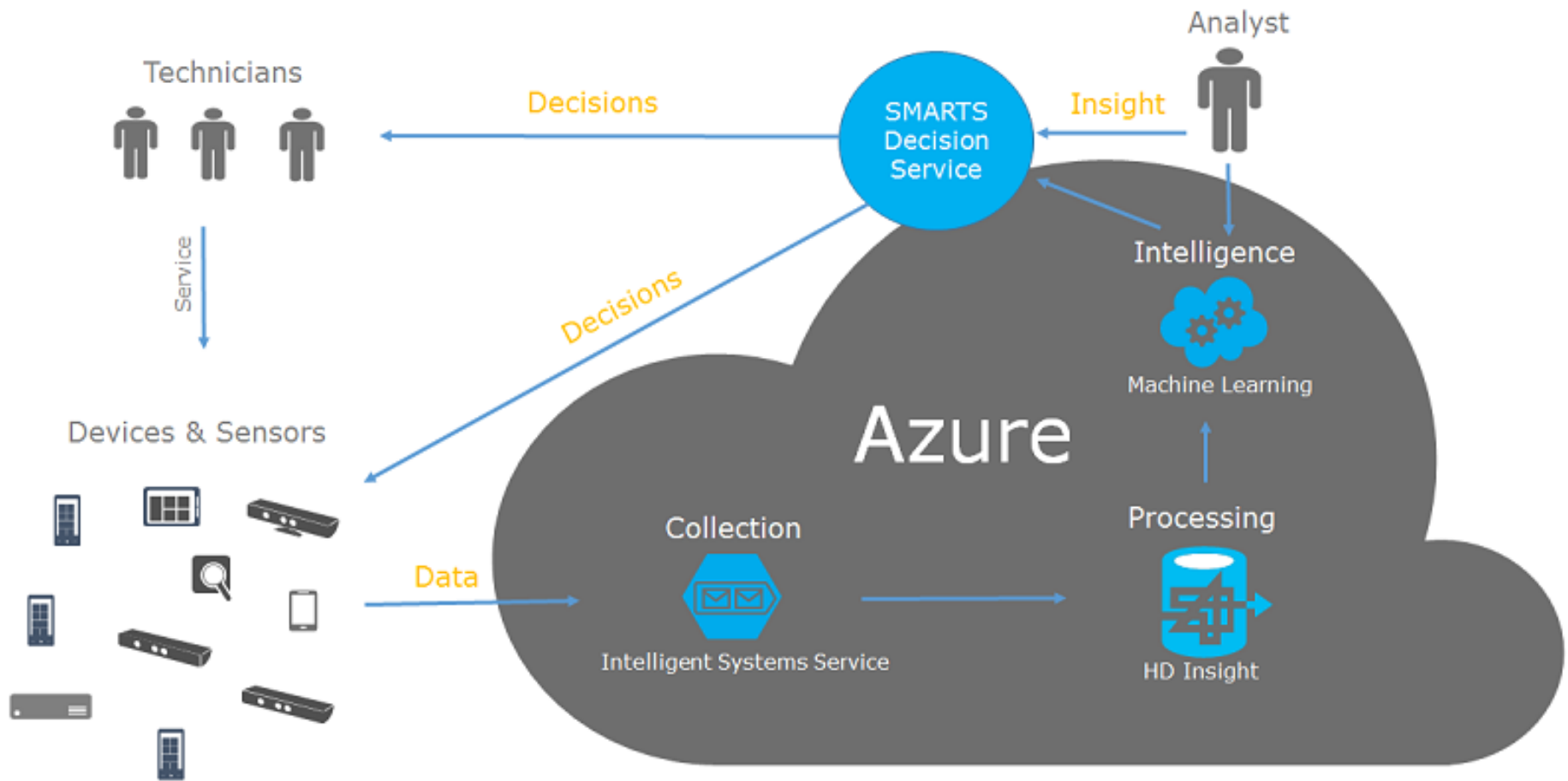
Machine Learning as a Service (MLaaS)



Machine Learning “as a Service” (MLaaS)

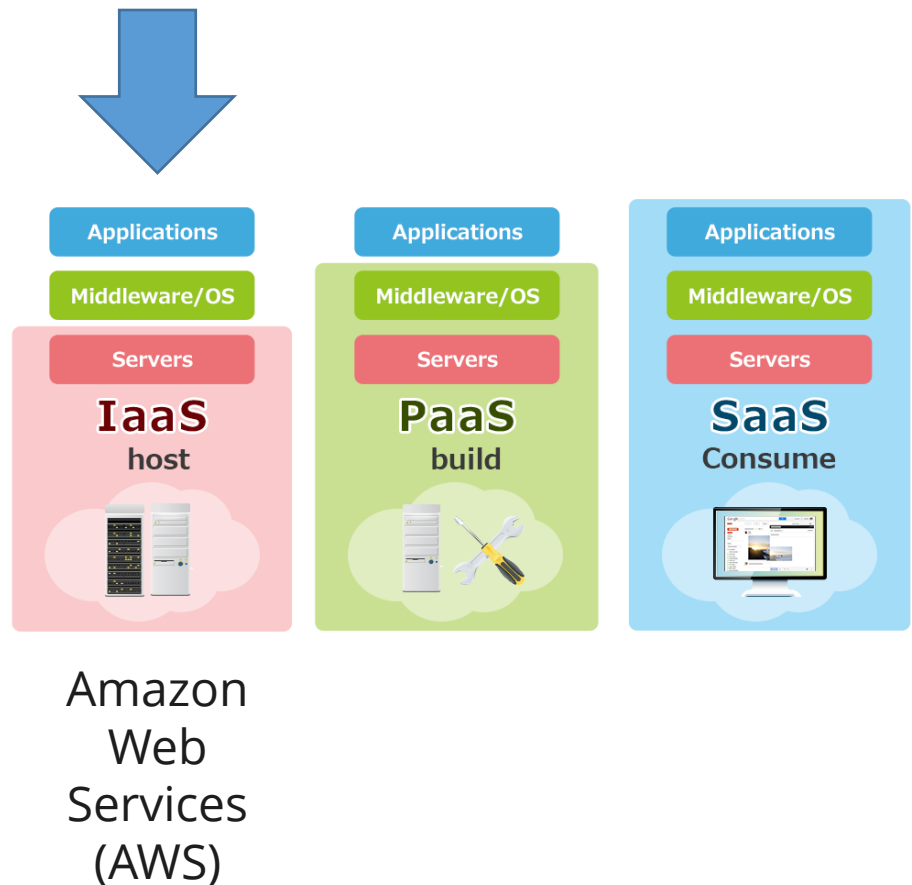
- **Machine Learning as a Service** is an umbrella definition of automated and semi-automated cloud platforms that cover most infrastructure task such as
 - Data pre-processing
 - Model training
 - Model evaluation
 - Dataset management
- Prediction results can be bridged for example with your internal IT infrastructure through REST APIs.
 - **REST (Representational State Transfer) API** is a set of rules that allows different software applications to communicate over the web using standard HTTP methods like GET, POST, PUT, and DELETE.

An example: the MS Azure



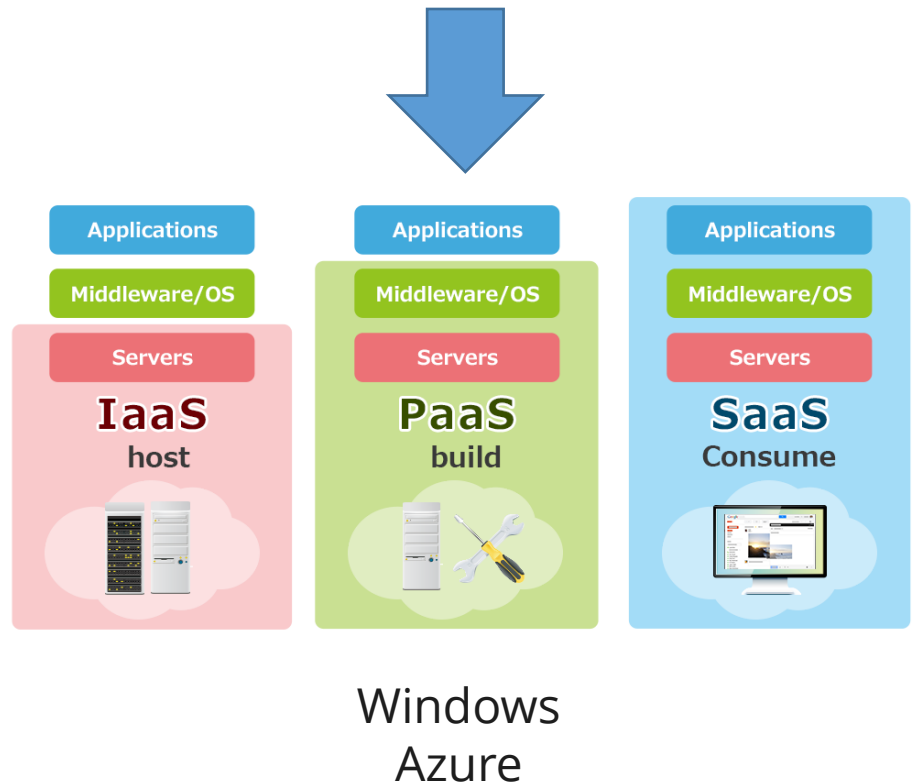
Cloud services: the 3-Tiers model: IaaS

- **Infrastructure as a service (IaaS)**
- A vendor provides clients pay-as-you-go access to
 - storage,
 - networking,
 - servers and
 - other computing resources in the cloud



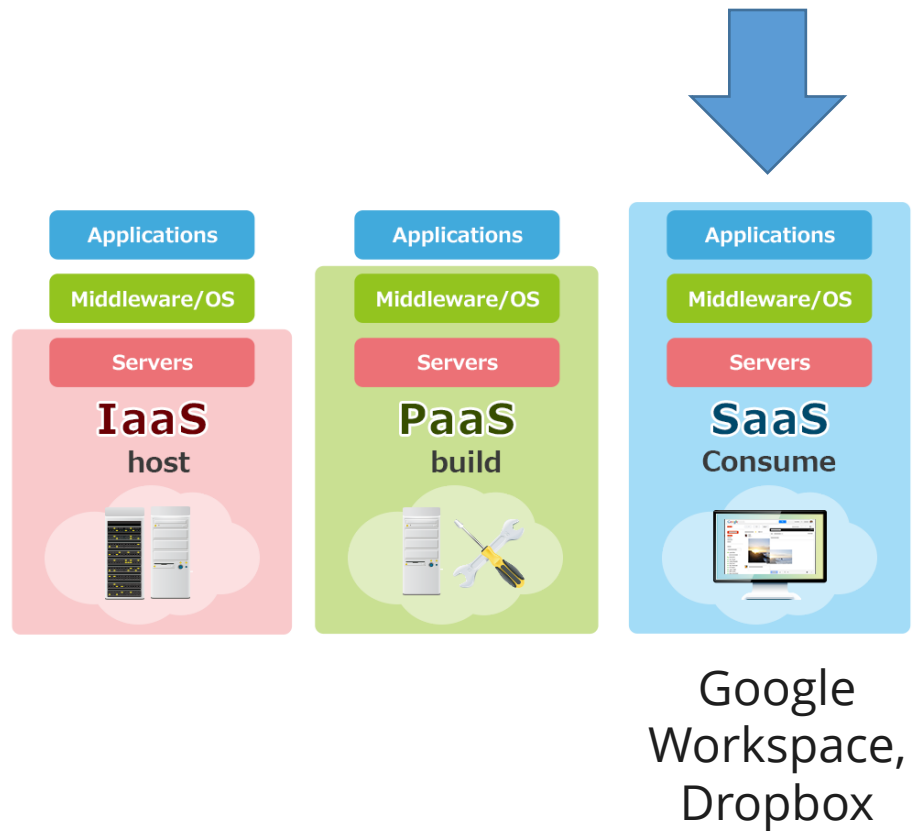
Cloud services: the 3-Tiers model: PaaS

- **Platform as a service (PaaS)**
- A service provider offers access to a cloud-based environment in which users can build and deliver applications.
- The provider supplies underlying infrastructure.



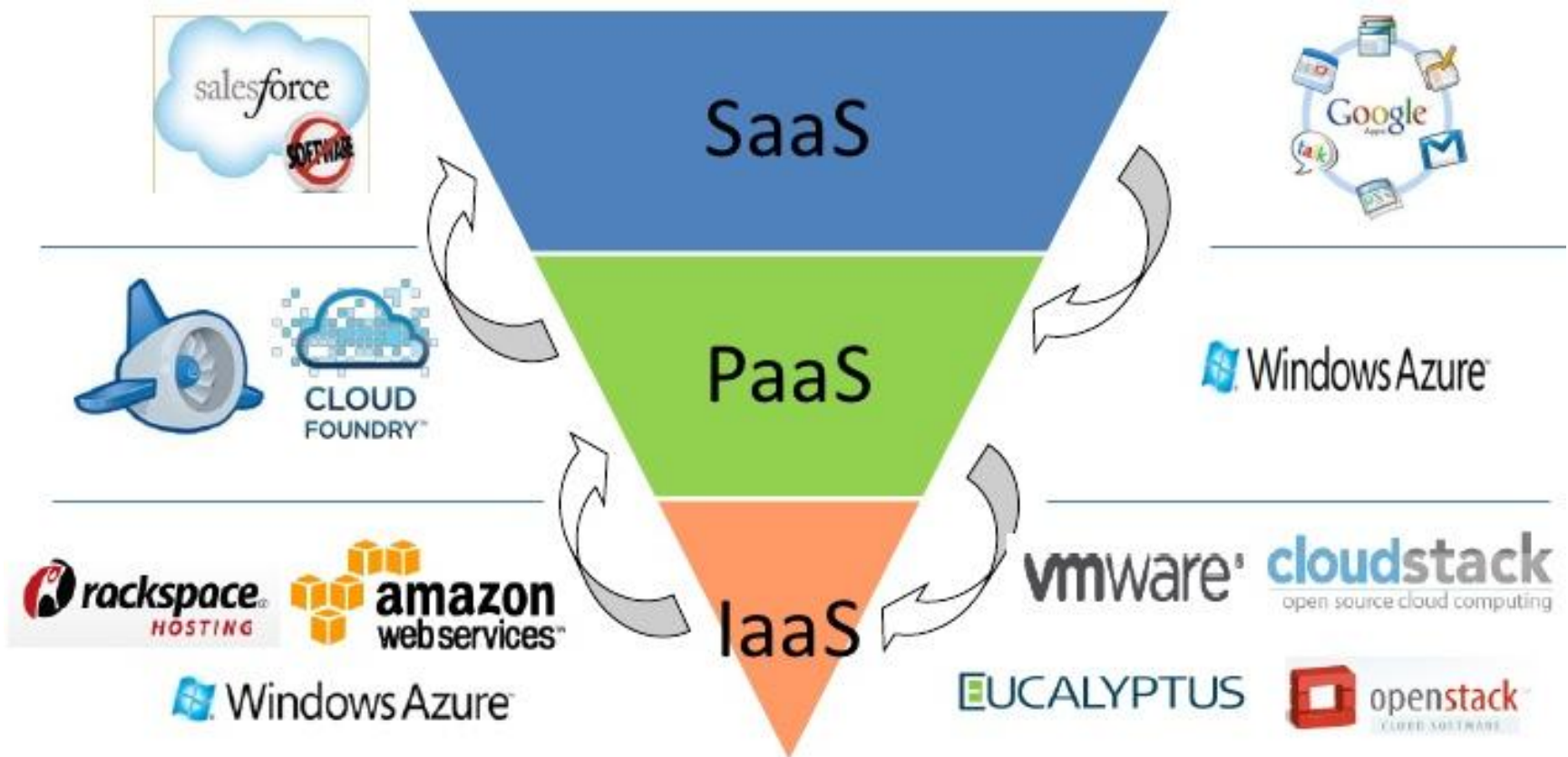
Cloud services: the 3-Tiers model: SaaS

- **Software as a service (SaaS)**
- A service provider delivers software and applications through the internet.
- Users subscribe to the software and access it via the web or vendor APIs.



Cloud services: the 3-Tiers model

Note: Things are changing quickly ...



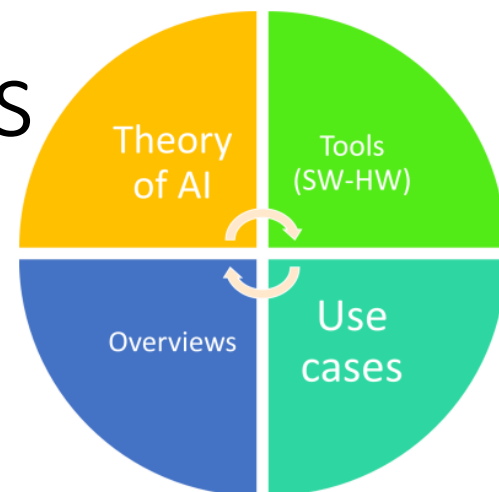
3 leading cloud MLaaS services

- Product names, interfaces, and services evolve over time. Check the websites of the main cloud providers for their current MLaaS and AI platform offerings:
 - Google Cloud
 - AWS
 - Microsoft Azure
- Focus on the design concepts, not on memorizing product names and specs.

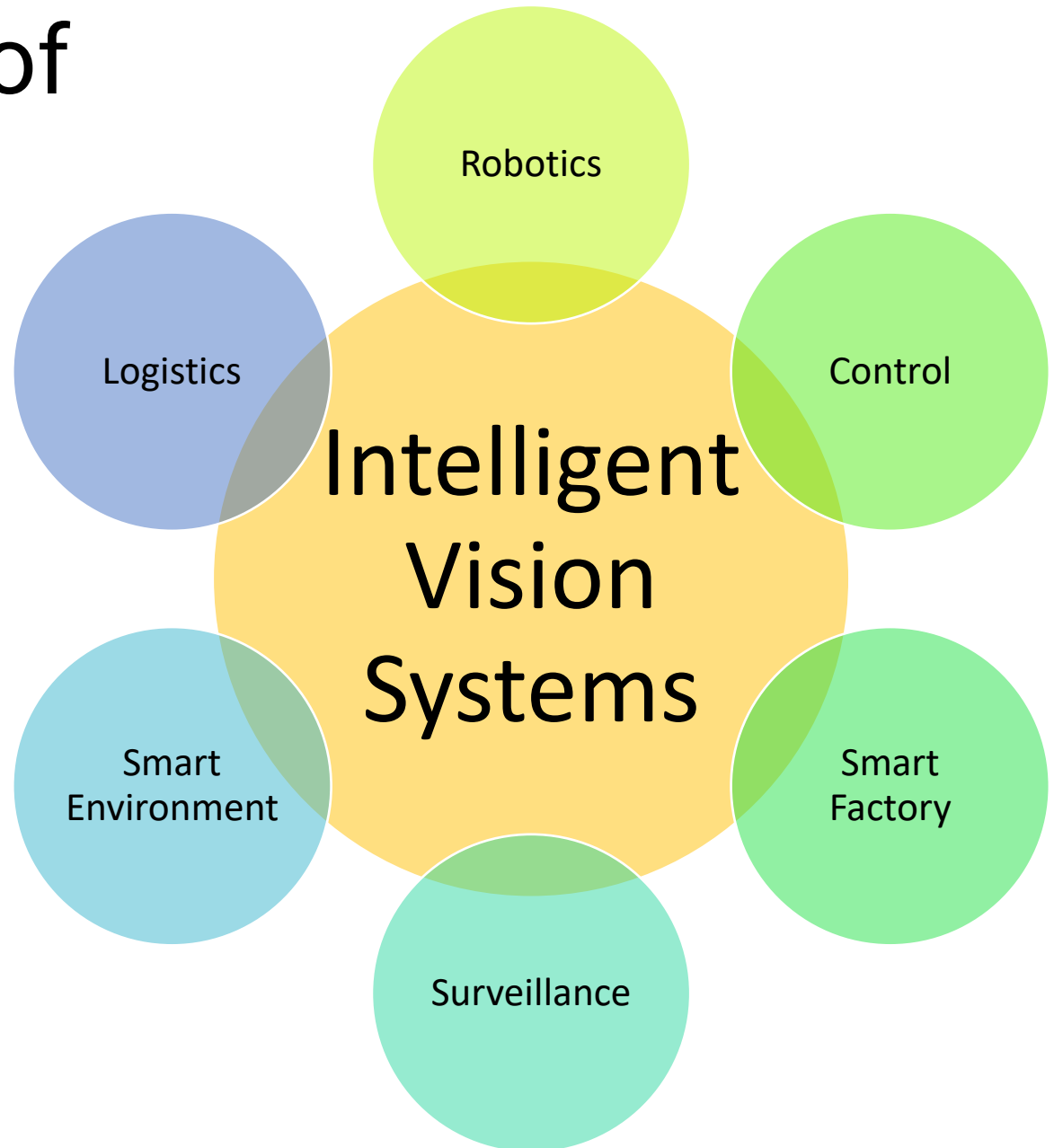


Overview

Introduction to Intelligent Vision Systems



6 examples of Intelligent Vision Systems



Intelligent systems for Robotics



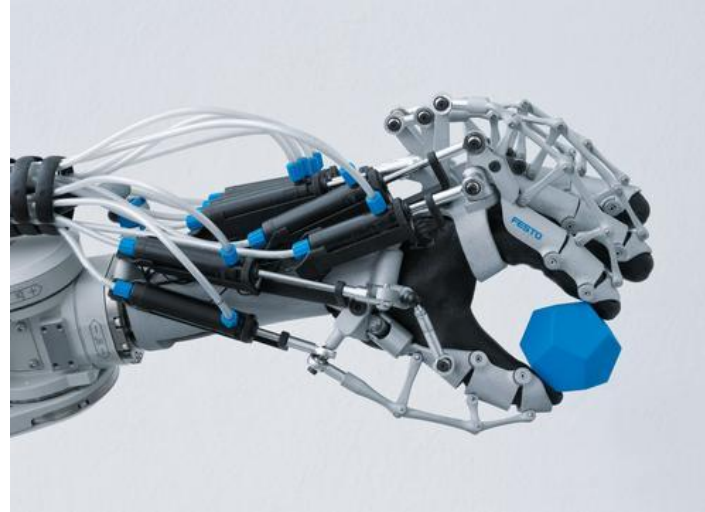
- Automatic machines that perform various tasks that were previously done by humans
- Example:
 - Industrial robots (pick and place, cutting, welding)
 - Pilot-less combat airplanes
 - Land-mine hunters
 - Autonomous vacuum-cleaners
- Components: Body structure, actuators, power-source, sensors, controller (the AI-based part)

Intelligent systems for Robotics (2)

General purpose



Industrial applications



Trajectories optimization, haptics, sensors, improved controllers, vision systems, co-working, ...

Many of them are using cameras



Methods and applications: **Control**



Control of a plant

Applications of intelligent system for control of industrial processes, industrial automation, robotic systems, complex products, power distribution grids, automotive and transport systems.

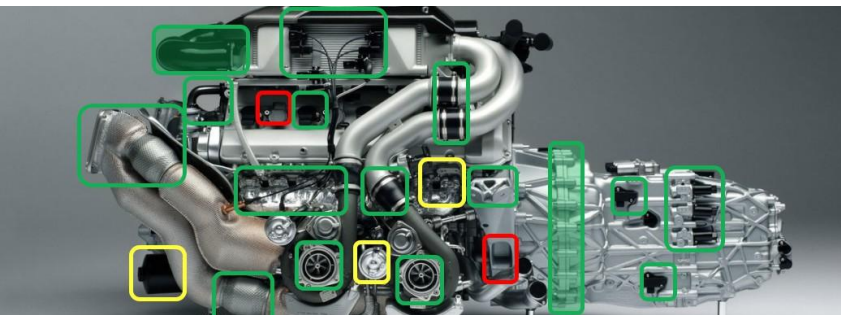
Control of a position



Stability Control



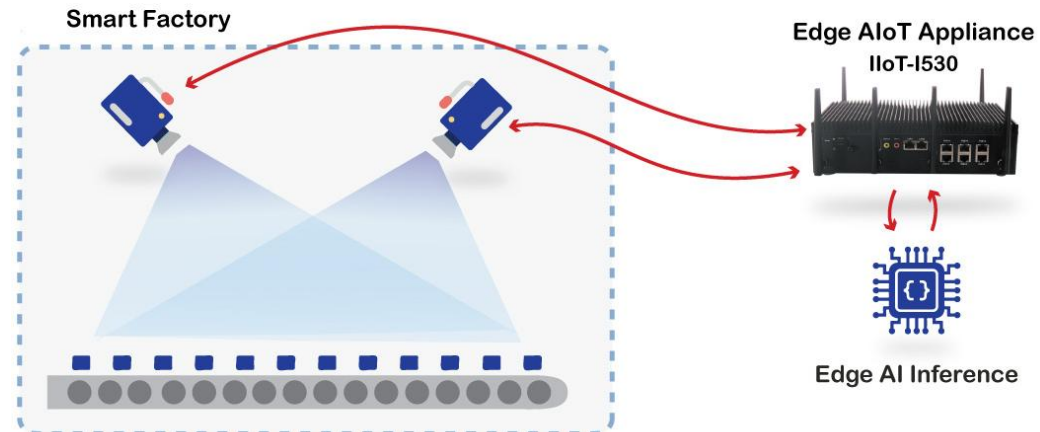
Intelligent Vision Systems for Production Inspection (smart factory)



Non ideal inputs and changes in time



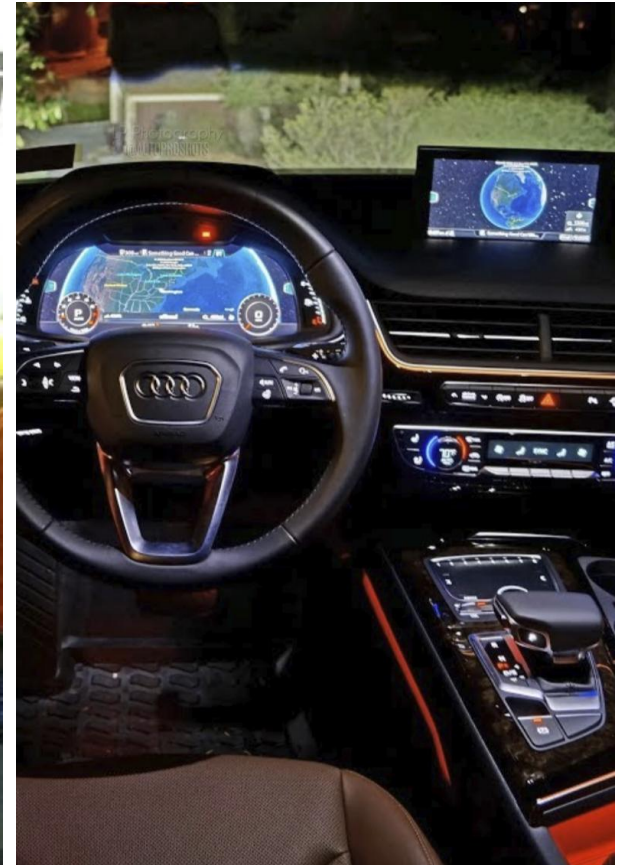
Machine Vision-based Factory Product Inspection



Intelligent Vision Systems for surveillance and security

Complex and different scenarios to be managed





Smart Environments: Cars

- Identification of the driver
- Driving assistance
- Autonomous driving
- Security/Surveillance
- Automatic settings ...



Smart Environments: Entertainment

- Identification
- Profiling (adult/kid/dog/cat)
- Security
- ...



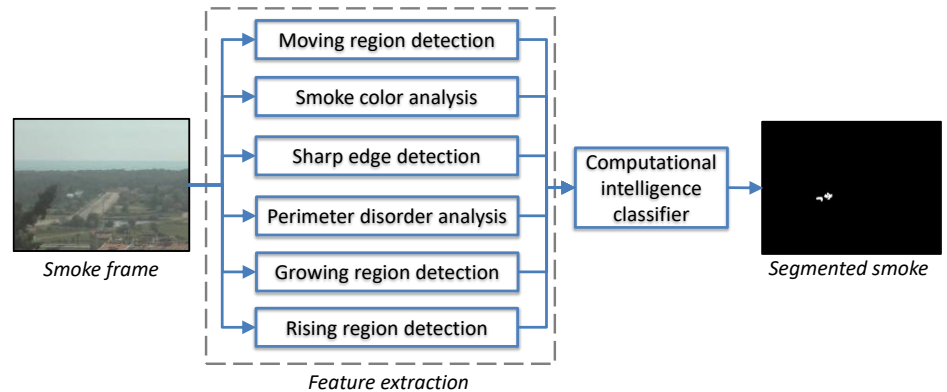
Smart city

- Intelligent Traffic Management
- Anomaly detection
- Crime prevention
- Security/Surveillance
- Profiling (adult/kid)
- ...

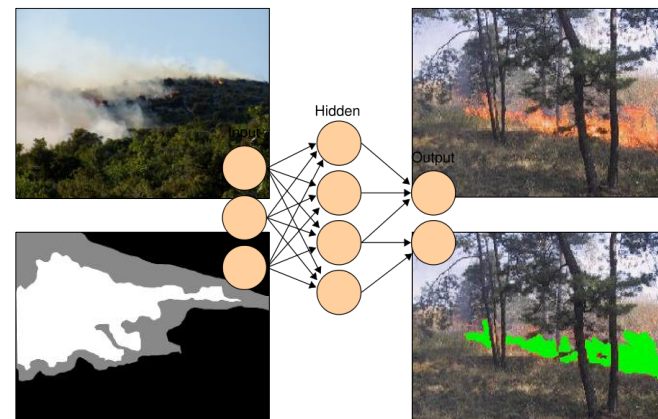
Environmental applications: Wildfire smoke detection

- Problem in regions with hot climate and extensive vegetation
- Current techniques require large infrastructure and human inspection
- Innovative computer vision approach
 - Visible light
 - Low quality frames
 - Limited hardware power
 - Different environments
 - Distinctive features:
 - Static
 - Dynamic
 - Computational intelligence
- Flexible: can learn new environments!

STEP 1: Feature extraction (by the designer)

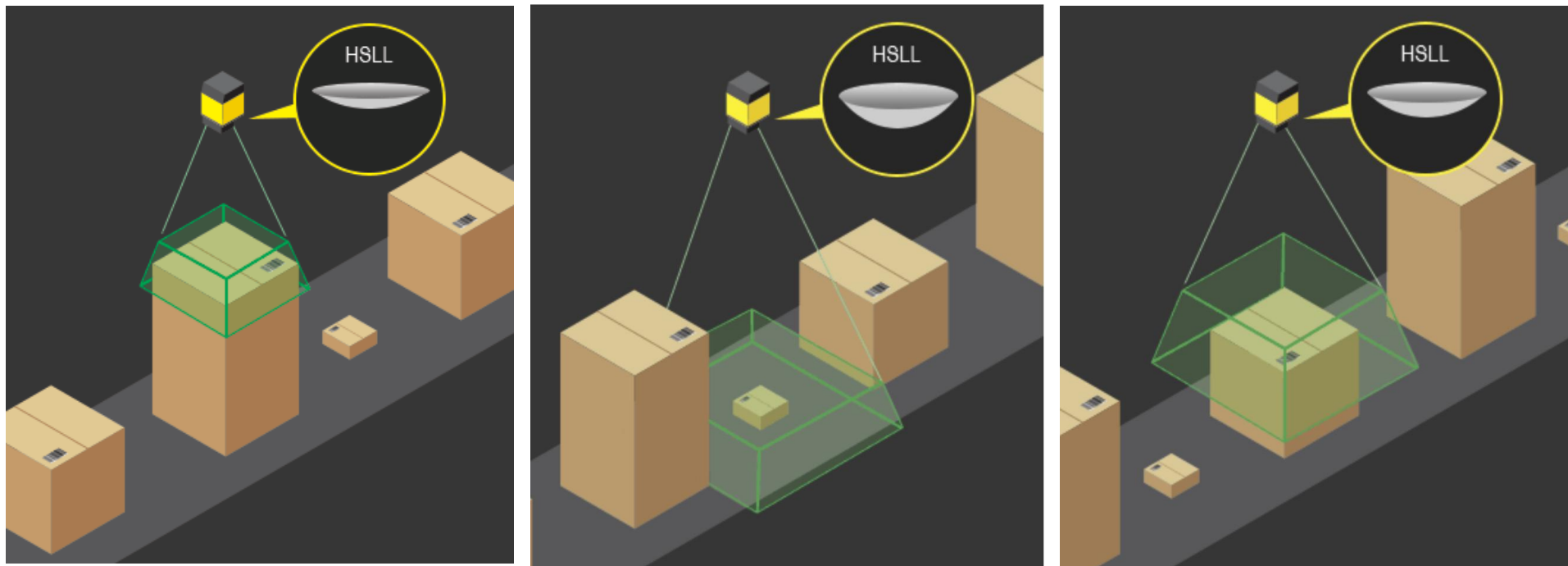


STEP 2: classification with NN



AI in **Logistics** (+liquid lenses)

- On a conveyor belt the bar codes can be effectively scanned by changing the focal of the optics
+ Deep Learning OCR → correct identification



Methods and applications: Logistics

SEEING, SPEAKING & THINKING LOGISTICS OPERATIONS



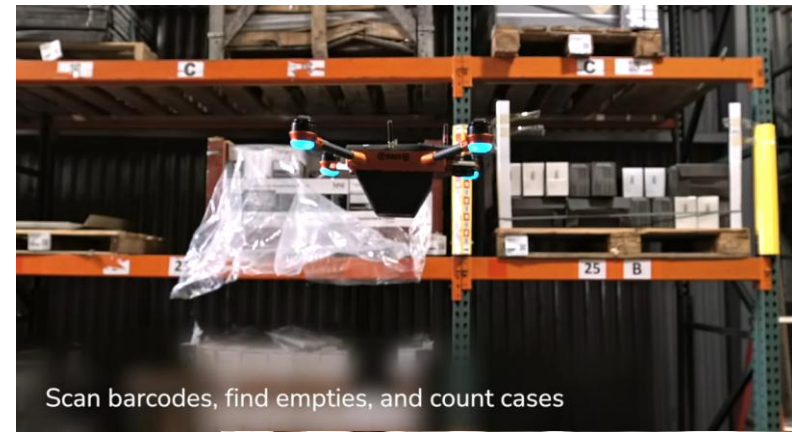
Drone inventory

- Drone-based Inventory Automation: a \$1.9 Trillion Warehousing Industry
- Corvus Robotics, founded by Mohammed Kabir '21, is using drones that can navigate in GPS-denied environments to expedite inventory management.

<https://www.corvus-robotics.com/corvus-one>



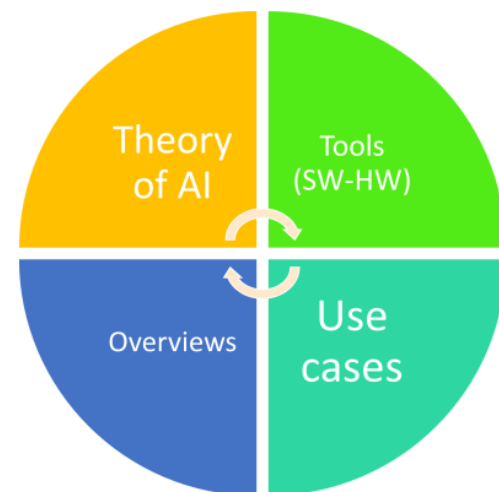
<https://youtu.be/C-gO7dCocV8>





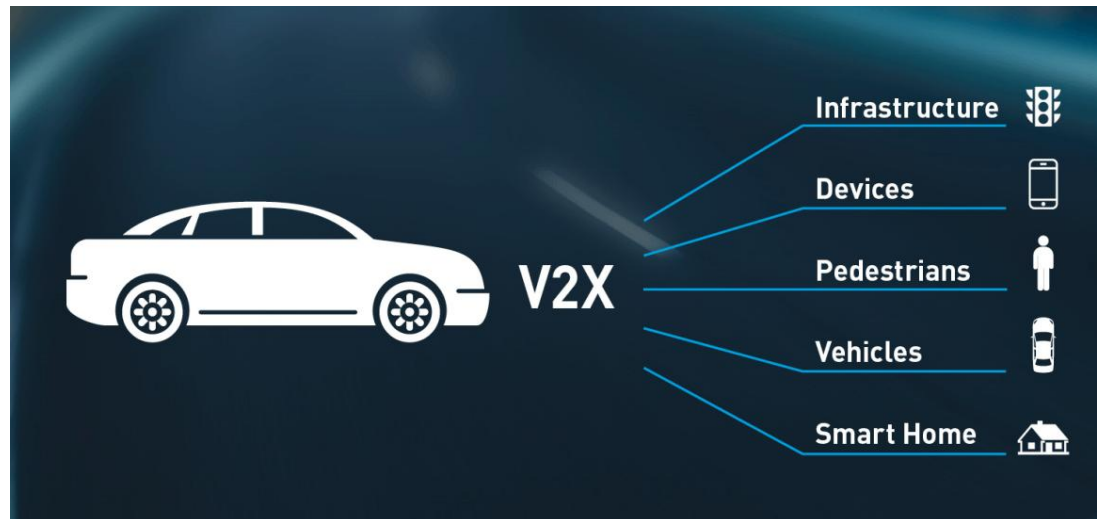
Overview Intelligent transportations

Current application and trends



Vehicle to everything (V2X)

- Vehicle to everything (V2X) is a term that refers to
 - high-bandwidth
 - low latency
 - highly reliable communication between a broad range of transport and traffic-related sensors.
- 5G mobile networks will be key to providing connectivity for vehicle to vehicle (V2V) and vehicle to infrastructure (V2I)



Road map for V2X

(think to electric/hybrid cars)

V2V - Vehicle-to-Vehicle.

Alerts one vehicle to the presence of another. Cars “talk” using DSRC technology.

V2D - Vehicle-to-Device.

Vehicles communicate with cyclists' V2D device and vice versa.

V2P - Vehicle-to-Pedestrian.

Car communication with pedestrian with approaching alerts and vice versa.

V2H - Vehicle-to-Home.

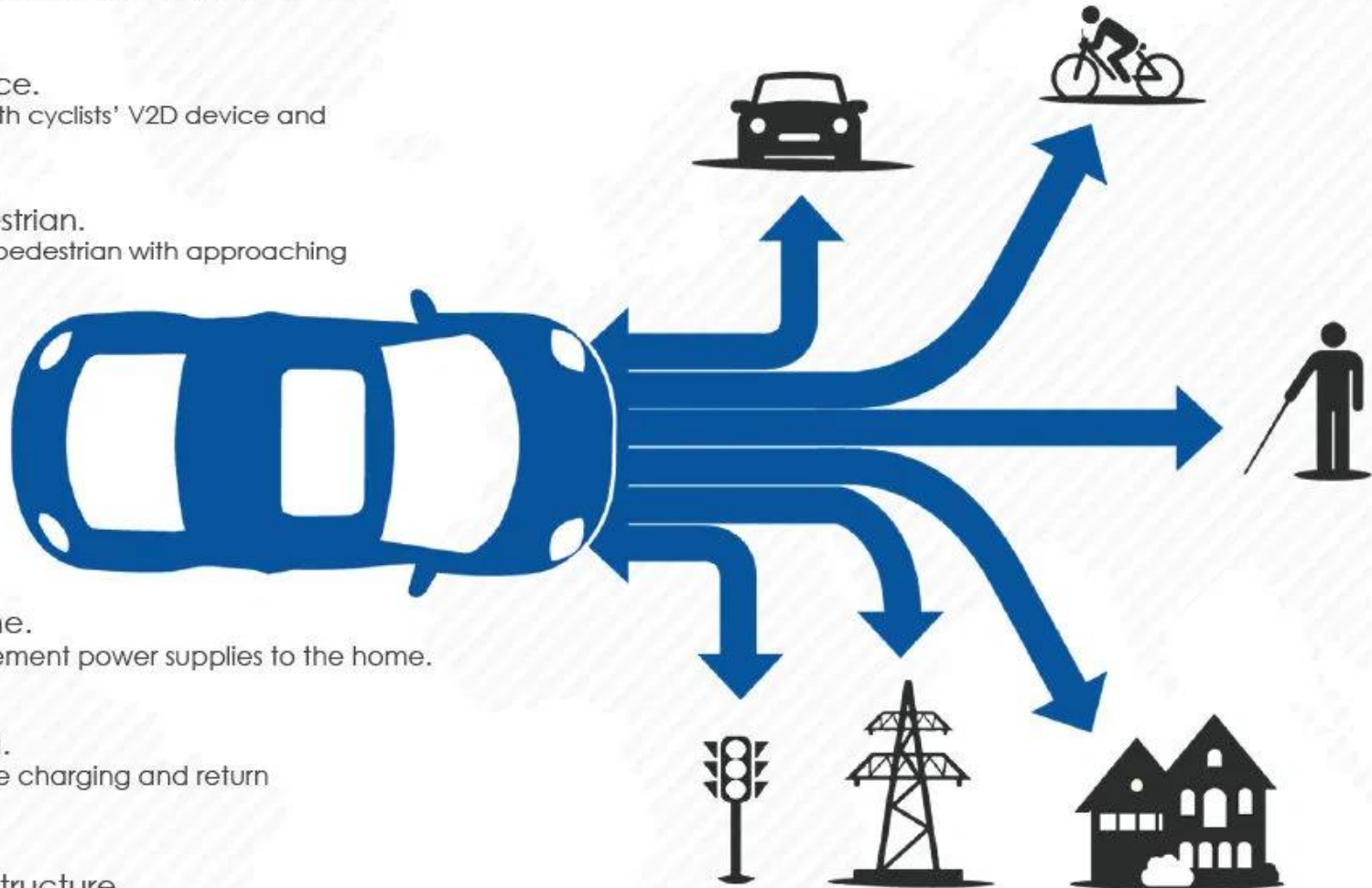
Vehicles will act as supplement power supplies to the home.

V2G - Vehicle-to-Grid.

Smart grid controls vehicle charging and return electricity to the grid.

V2I - Vehicle-to-Infrastructure.

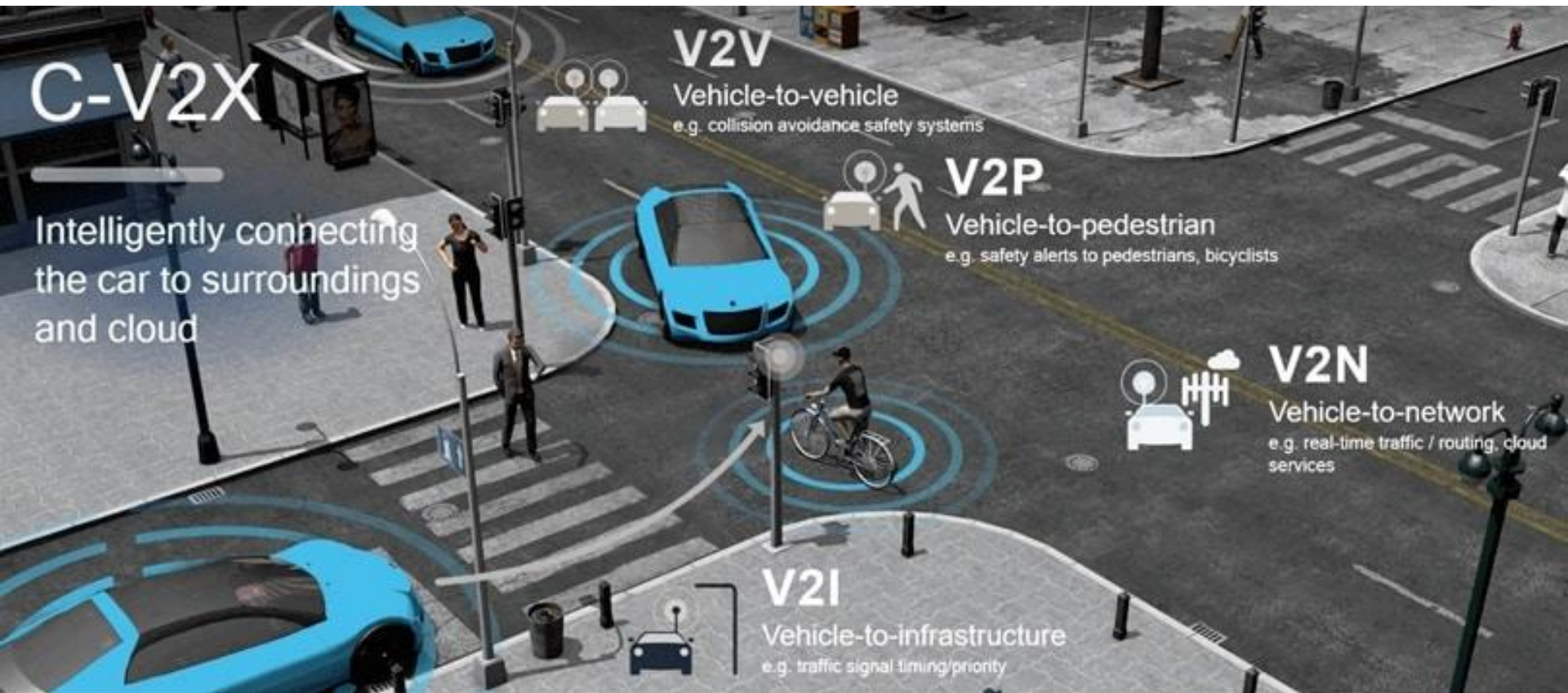
Alerts vehicles to traffic lights, traffic congestion, road conditions, etc.



Functions of V2X

Image to be a pedestrian approaching a zebra crossing: traffic lights can switch color; cars will be warned before.

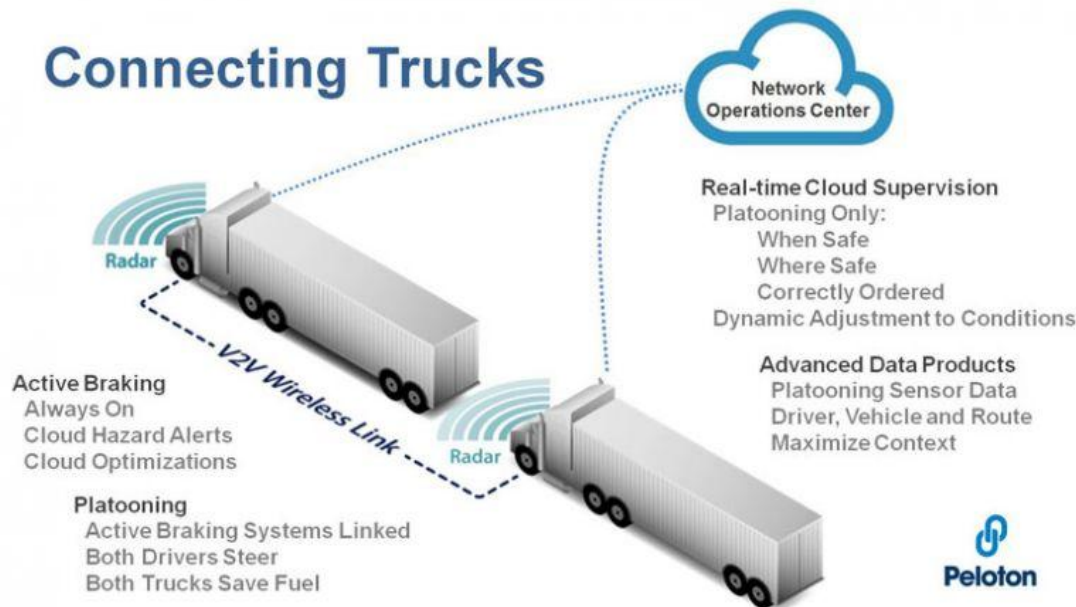
→ More fluid and safe circulation



IoT: Platooning vehicles



- **IoT-enabled platooning** is an innovative transport system where trucks can drive closely together – one after another – using a common communication system based on V2V IoT



Tesla, Volvo, Iveco, Scania, ...
are prototyping
Market introduction 2023-30



Intelligent transportation systems



- Advanced systems for of transport and traffic management (highway, city, region level)
- Enabling users to be better informed and make safer, more coordinated, and 'smarter' use of transport networks using artificial intelligence
- Examples:
 - Calling for emergency services when an accident occurs
 - Using cameras to enforce traffic laws
 - Signs that mark speed limit changes depending on conditions and weather conditions
 - Collision avoidance systems



Intelligent transportation systems: key features

- **Wireless communications**

- Short-range communications (up to 350m using IEEE 802.11)
- Longer range communications using infrastructure networks such as WiMAX (IEEE 802.16), Global System for Mobile Communications (GSM), or 3G, 4G, 5G

- **Computational technologies**

- Micro controllers on sensors and car devices
 - IoT, dynamic data devices
- **Virtual sensors** (e.g., the ice presence sensors)

- **Heterogenous data fusion**

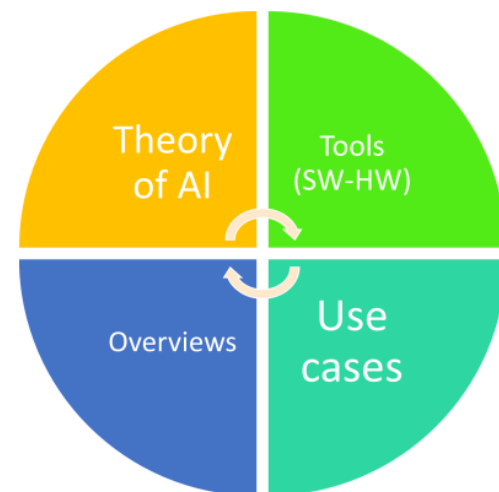
- From sensors, surveillance cameras, smartphones, GSM towers
- Different protocols and artificial intelligence!



Use cases

The Pirelli Cyber (aka Connesso)

From a sensor to the cloud computing



From a small sensor → smart city



PIRELLI DESIGNS THE MOBILITY OF THE FUTURE IN 5G

SMART CYBER TYRES COMMUNICATE POTENTIALLY DANGEROUS ROAD CONDITIONS TO THE VEHICLE AND TO THE INFRASTRUCTURE INTEGRATING INSIDE A V2X COMMUNICATION

STAGE 1

INFORMATION COMMUNICATION

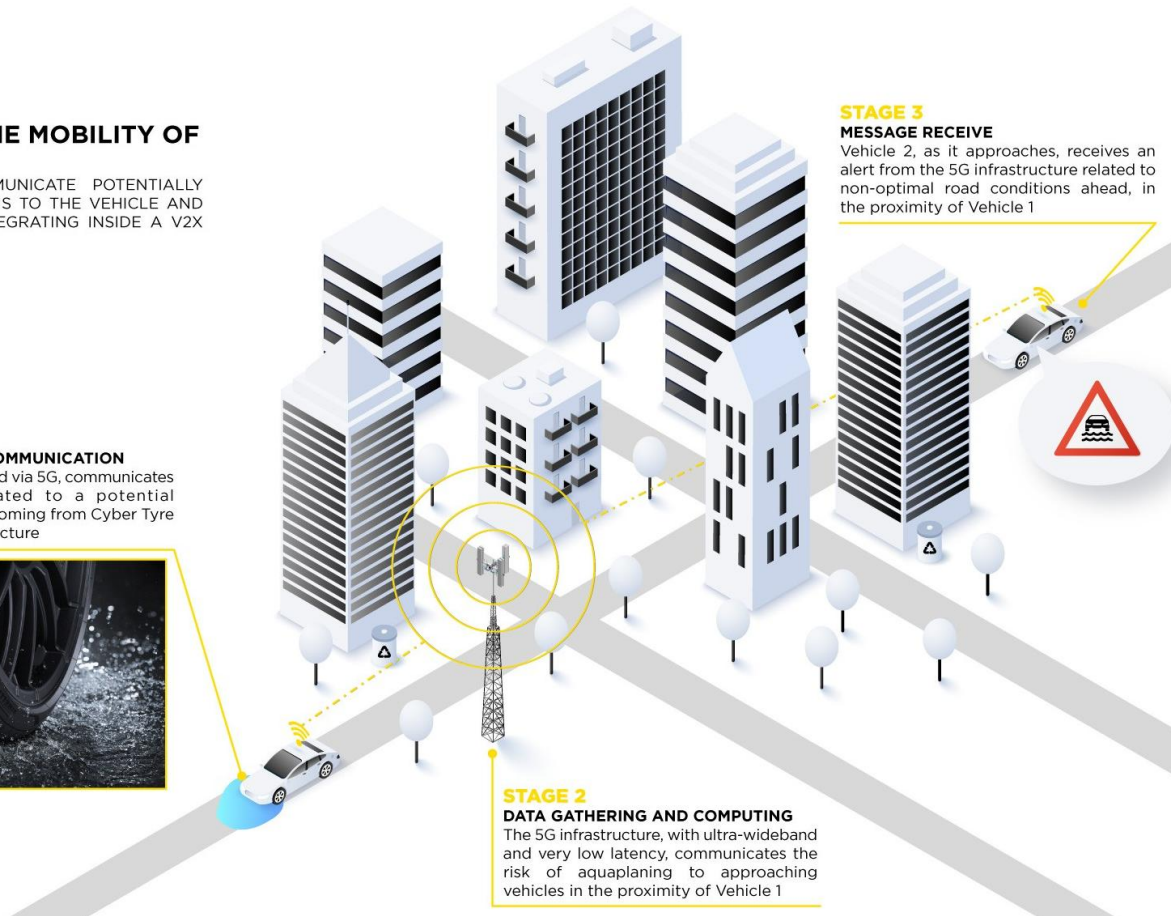
Vehicle 1, connected via 5G, communicates information related to a potential aquaplaning risk coming from Cyber Tyre to the 5G infrastructure



STAGE 3

MESSAGE RECEIVE

Vehicle 2, as it approaches, receives an alert from the 5G infrastructure related to non-optimal road conditions ahead, in the proximity of Vehicle 1

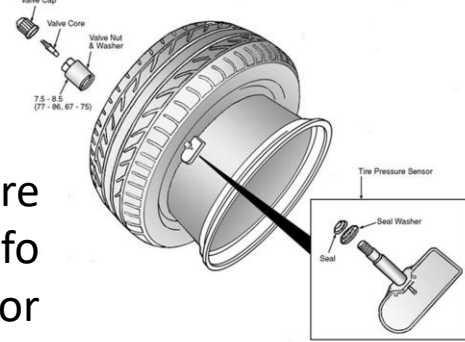


The tyre sensors are in contact with the road

- The tyre sensors detect potential danger on road surfaces (water, poor grip, ...)
- The car adapts its control and driving assistance systems
- The car provides the same information to other cars and the infrastructure

The Pirelli CYBER CAR IoT

The tyre
pressure + info
sensor



PIRELLI CYBER CAR HOW DOES IT WORK

VEHICLE SETTINGS BASED
UPON TYRE DATA



VEHICLE



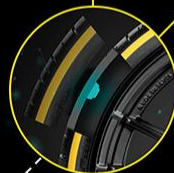
CLOUD

END USER

NOTIFICATION
MESSAGES



SENSOR INSIDE
THE TYRE



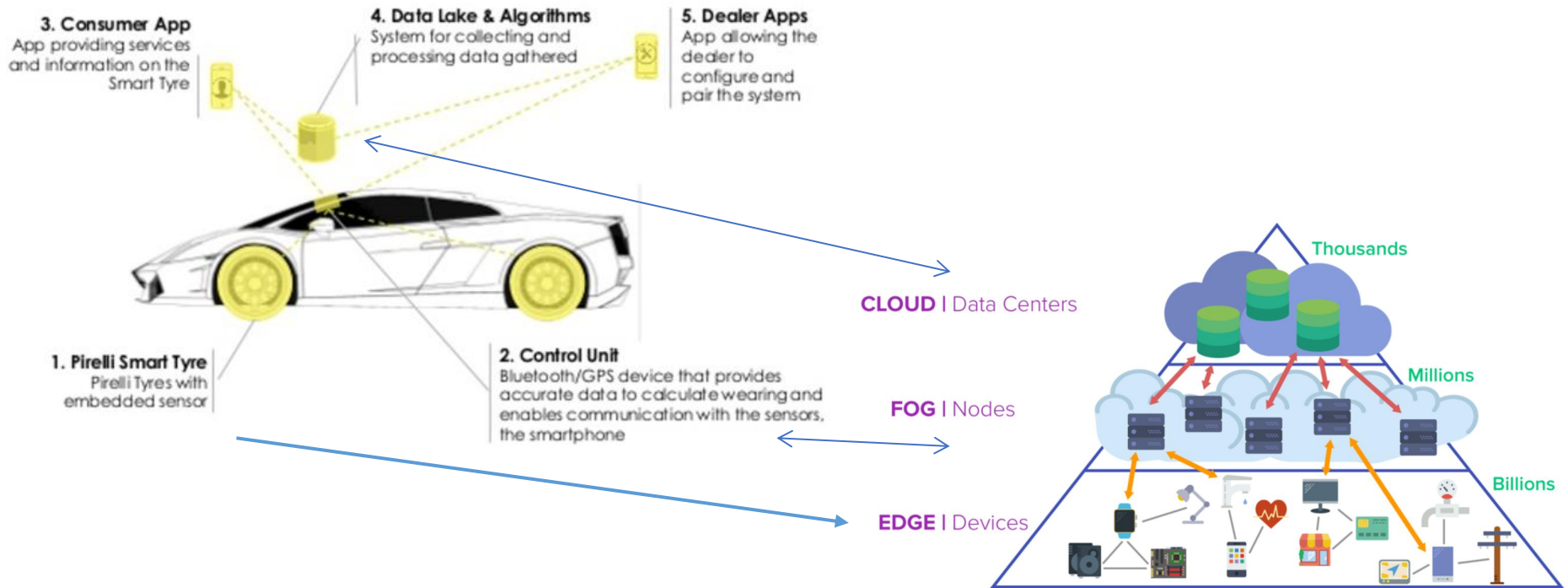
TYRE DATA GATHERED

- PRESSURE
- TEMPERATURE
- TYRE ID
- VERTICAL LOAD
- TYRE WEAR
- AUTOLOCATION



Pirelli Cyber Car: Edge $\leftrightarrow$ Fog $\leftrightarrow$ Cloud

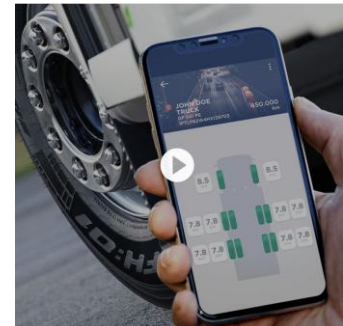
Amazon AWS



Pirelli Cyber: example of business opportunities from the Intelligent Sensor Architecture



<https://www.pirelli.com/tyres/en-au/car/tech-and-knowledge/cyber-tyre>
<https://www.pirellicyberfleet.com/>

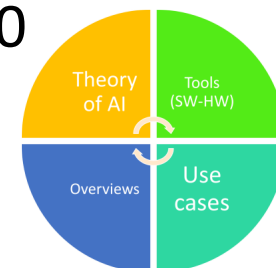
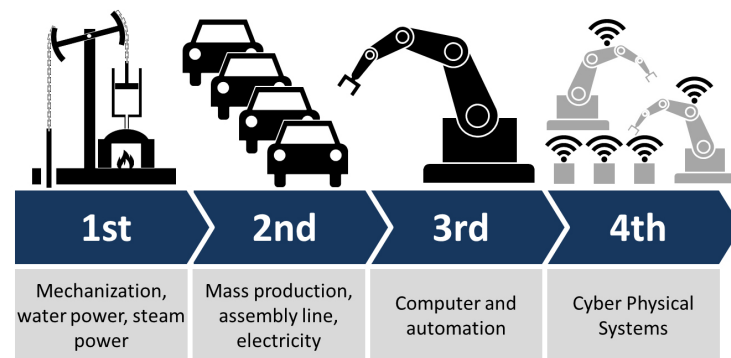


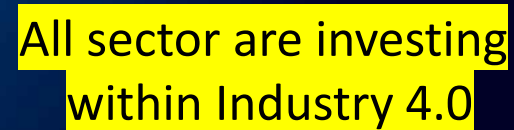


Overview

Industry 4.0

Features needed to create applications within Industry 4.0



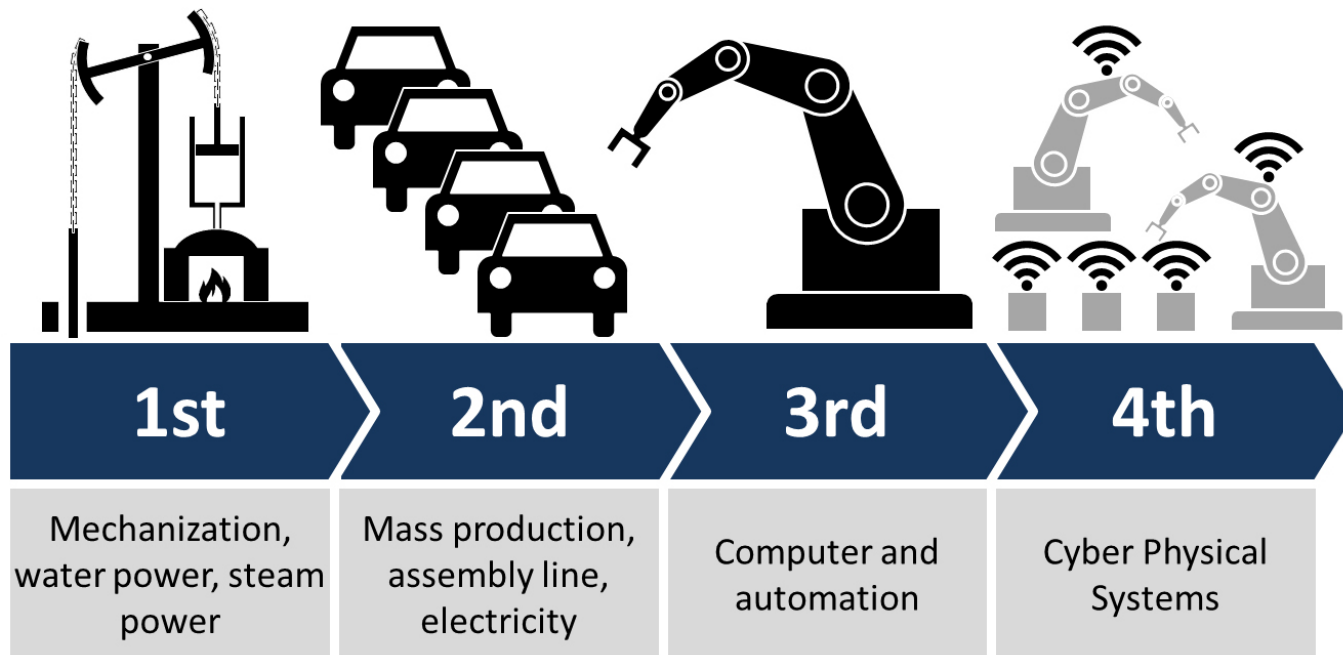


Many funding program within Industry 4.0 are available

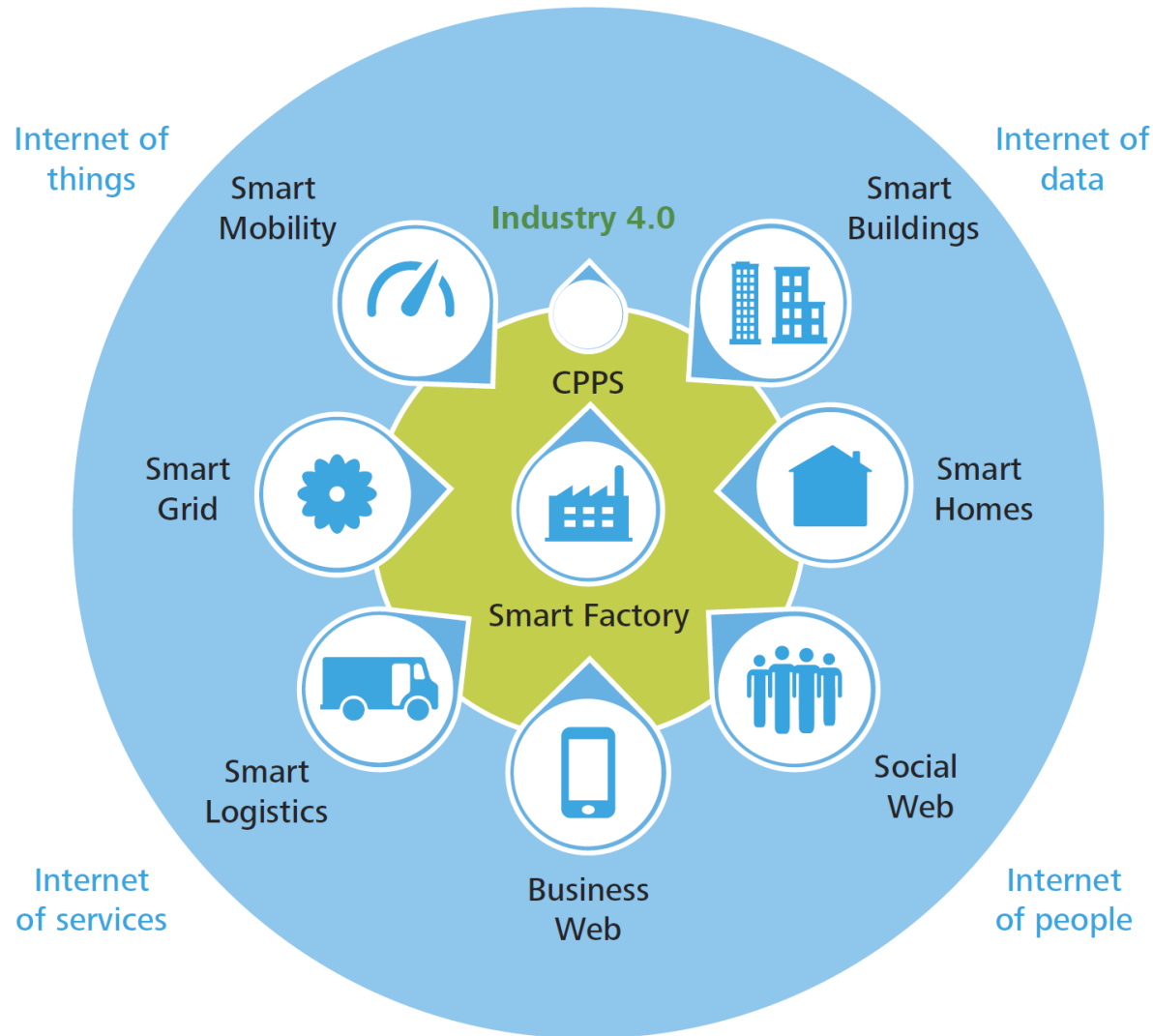
Industry 4.0

The topic of the course.....

- Stands for highly **intelligent** connected processes and **systems** created through digital networked data communication.

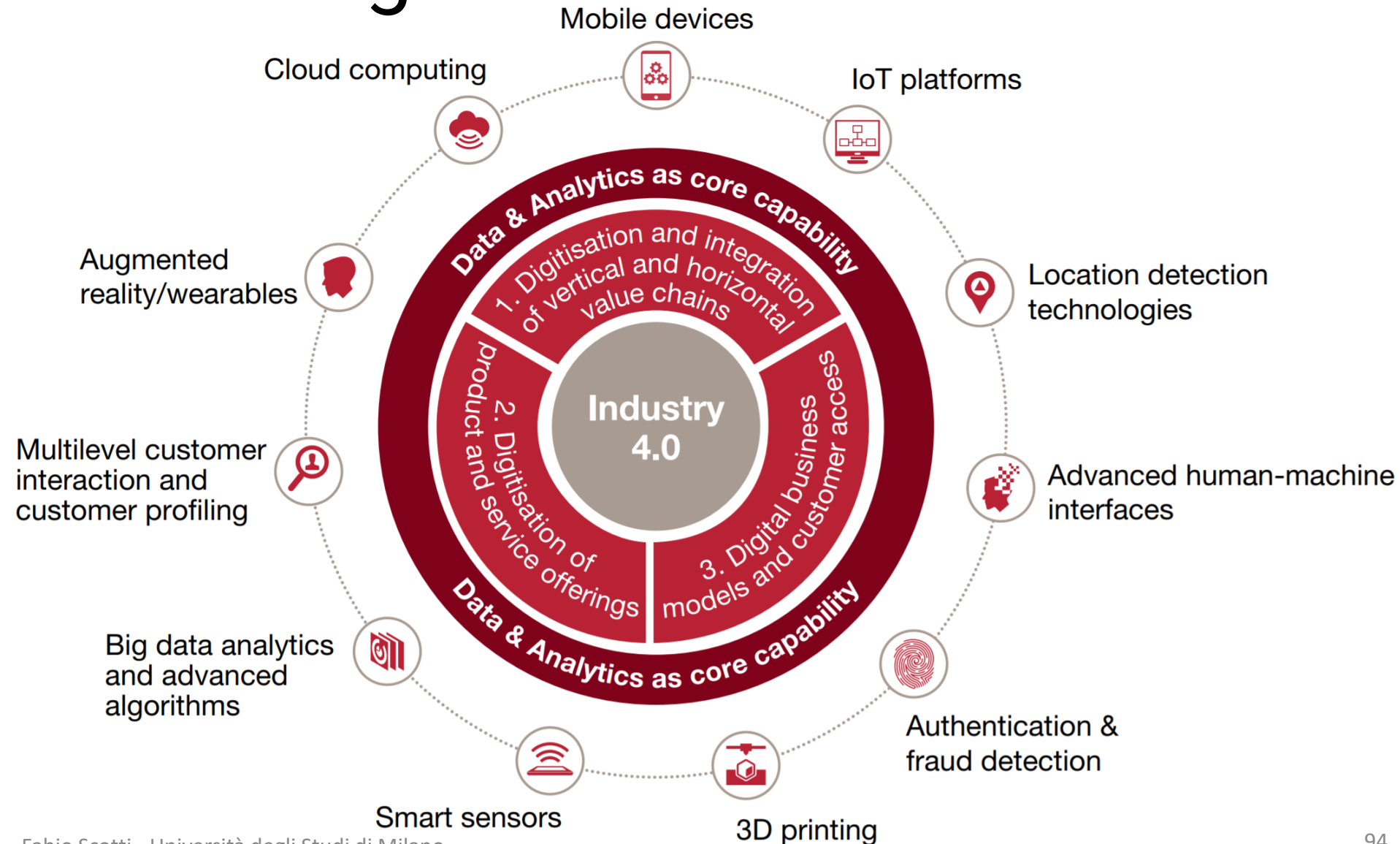


The I4.0 environment



(IoP) is the new infrastructure we need to create to enable people to communicate with each other using their own devices directly without relying on any trusted third party to establish the connection

14.0: Contributing digital technologies



Industry 4.0

Design principles (1-2)

- To be compliant to the Industry 4.0 funds...
- **Interoperability**:
 - The ability of machines, devices, sensors, and people to **connect and communicate with each other** via the Internet of Things (IoT) or the Internet of People (IoP).
- Information **transparency**:
 - The ability of information systems to **create a virtual copy of the physical world by enriching digital plant models with sensor data**. This requires the aggregation of raw sensor data to higher-value context information.

Industry 4.0

Design principles (3-4)

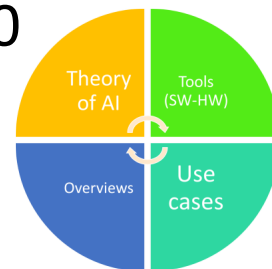
- Improved technical **assistance**:
 - The ability of assistance systems to **support humans by aggregating and visualizing information comprehensibly** for making informed decisions and solving urgent problems on short notice (exploiting the ability of cyber physical to conduct tasks that are unpleasant, too exhausting, or unsafe for their human co-workers).
- Decentralized **decisions**:
 - The ability of cyber physical systems to **make decisions on their own** and to perform their tasks as autonomously as possible.
 - Only in the case of exceptions, interferences, or conflicting goals, are tasks delegated to a higher level.



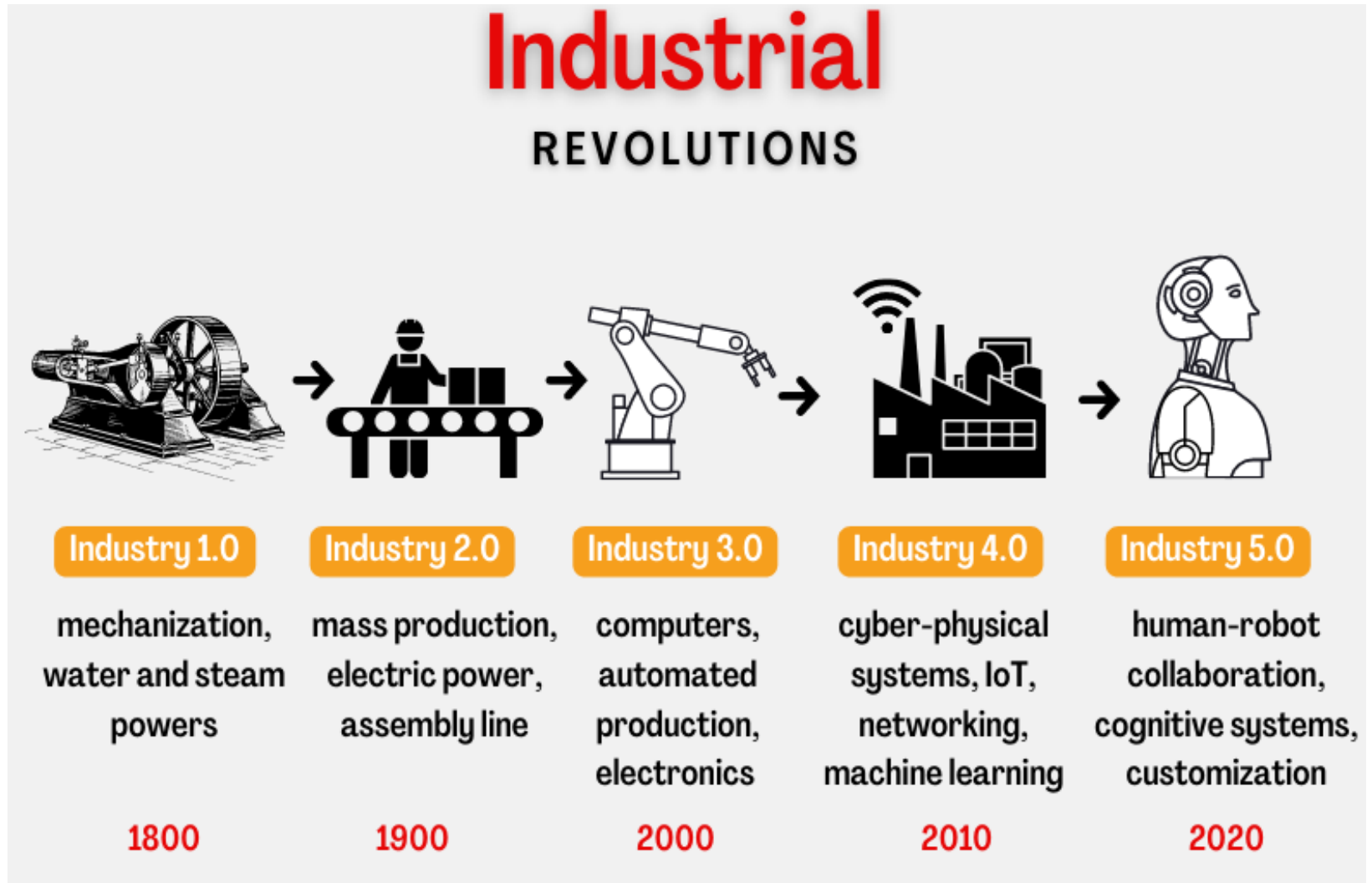
Overview

Industry 5.0

Features needed to create applications within Industry 5.0

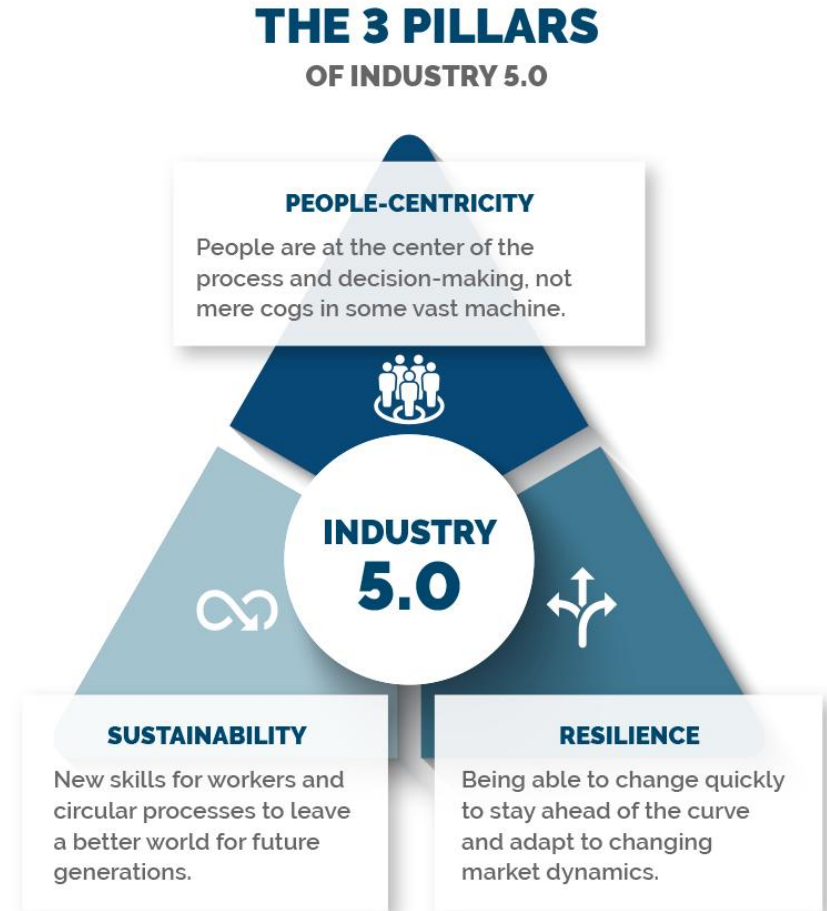


Industry 5.0



Industry 5.0

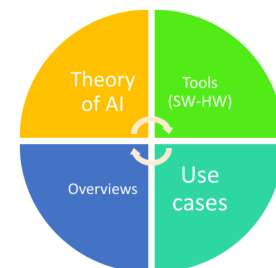
- adopting a **human-centric approach** for digital technologies including artificial intelligence
- **up-skilling** and re-skilling European workers, particularly digital skills
- modern, resource-efficient and sustainable industries and transition to a **circular economy**
- a globally **competitive and world-leading industry**, speeding up investment in research and innovation





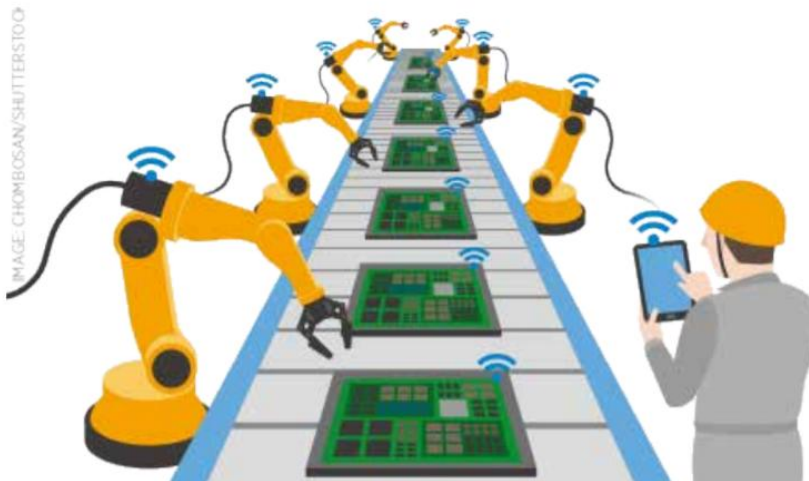
Overview Industrial Intelligent Vision Systems

Features needed to create applications within
Industry 4.0 and 5.0



Intelligent Vision System

- Industrial cameras continue to become smaller and more affordable, even as performance constantly improves.
- Machine Vision Imaging Systems, together with ever-expanding networking capabilities, opens up this potential for new applications and further advancing Industry 4.0.

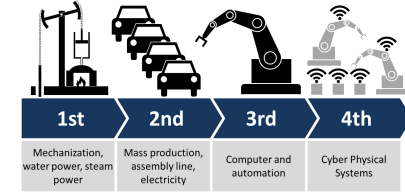


Cameras as tireless process auditors



Industrial cameras in production

Intelligent Vision System for Industry 4.0



- They are the **eyes** that capture the process and provide feedback via image processing software.
- Throughout all steps of manufacturing (from the inspection of raw materials and production monitoring to final inspections and quality assurance) industrial vision cameras are being used across all industries and **are an indispensable part of achieving efficiency and high quality standards (*)**.



History... today



2005-2006

- Machine Vision Cameras
- VGA to 3MP and Linescan
- Camera Link and FireWire Interfaces

2007-2008

- Smart Vision System
- DSP, Image Processing Library, Interactive GUI for Programming

2009-2013

- Smart Camera / Smart Sensor
- Embedded DSP
- Fully Integrated Functionality



2018-2020

- Smart cameras with FPGA
- With onboard deep neural network
- Measurements
- OCR + QR + Barcode adv. techniques

**Not training onboard
Just inference!**

Industrial vision systems

Sensors and smart cameras



3D laser
profilometer



Onboard
processing =
Smart camera

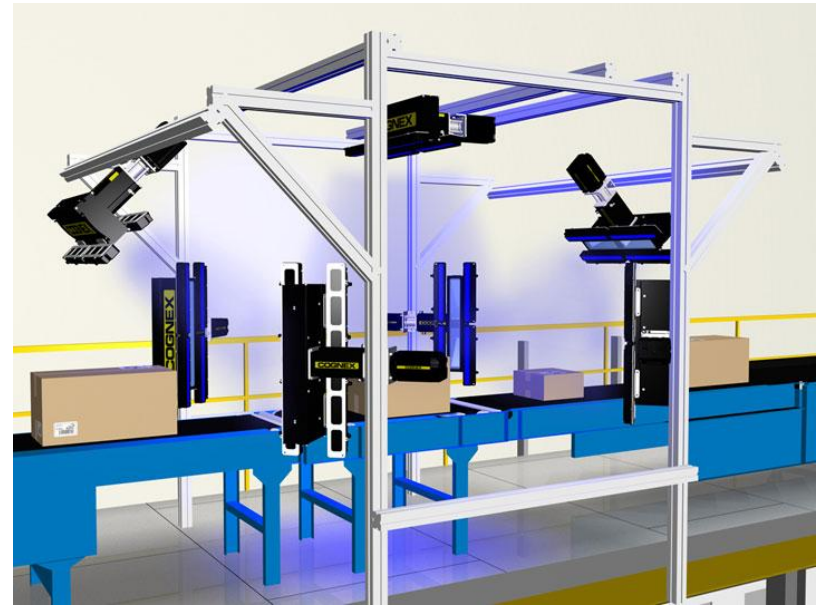
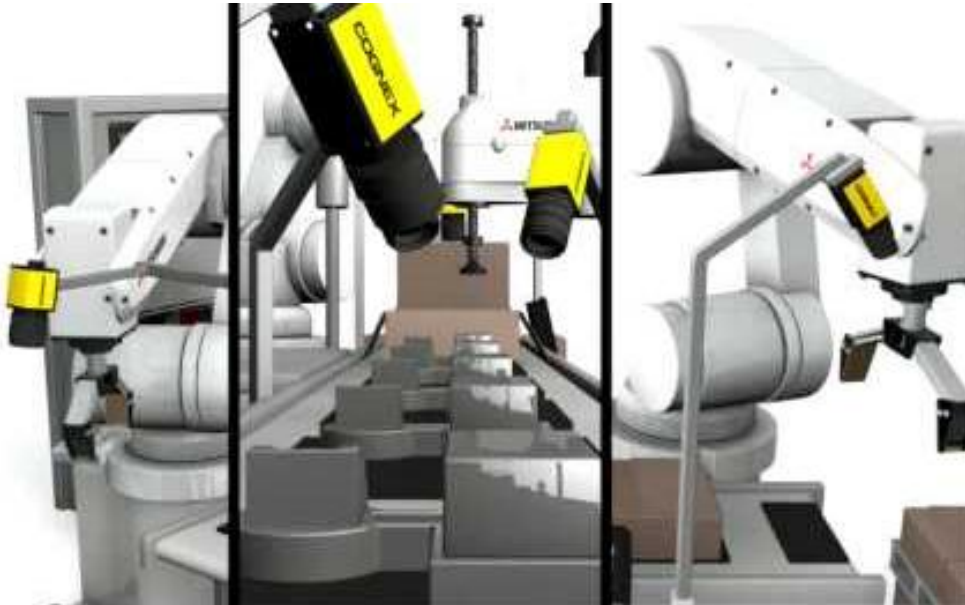


Camera



Industrial Vision Systems

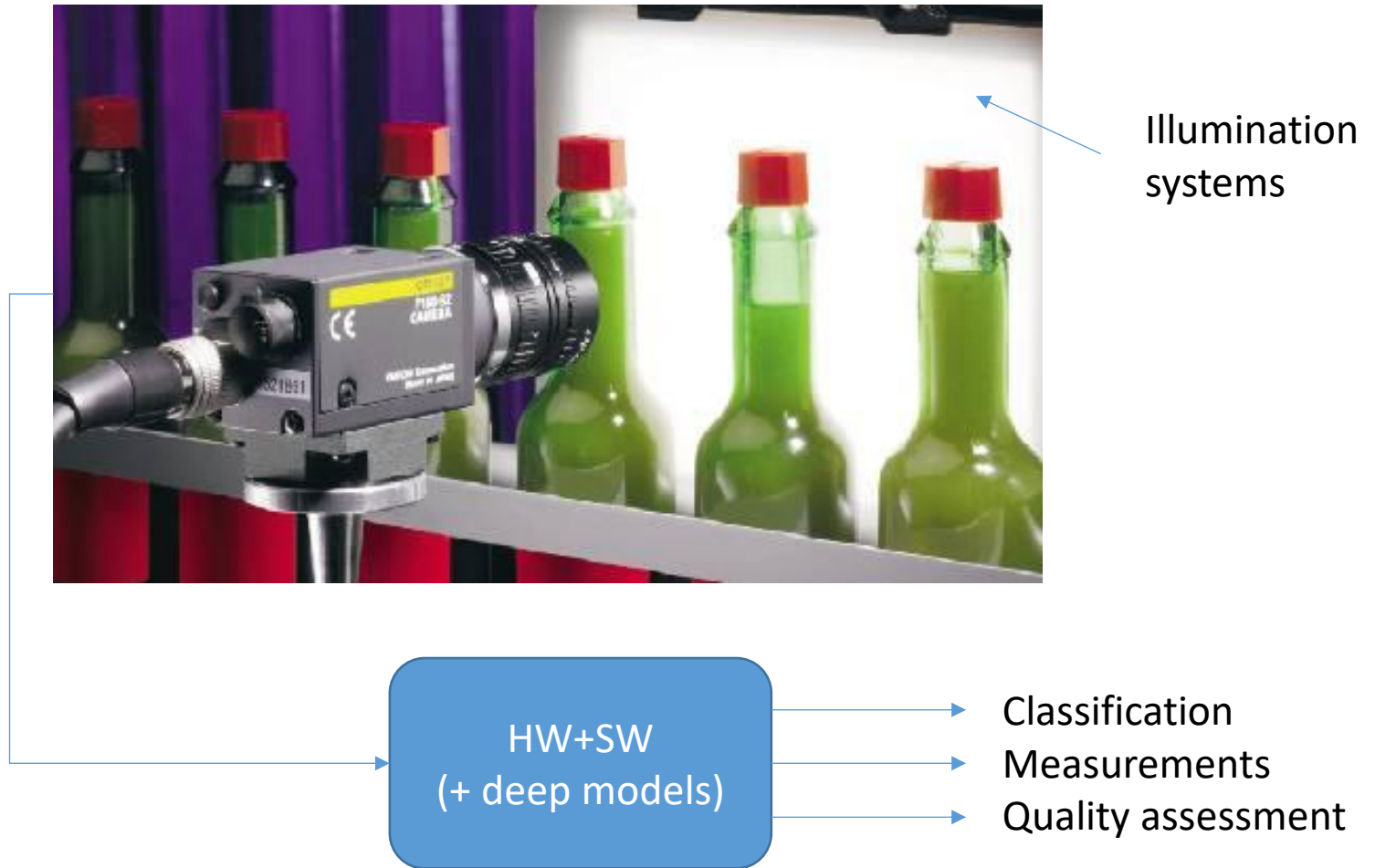
- Quality, Inspection, Production, Security



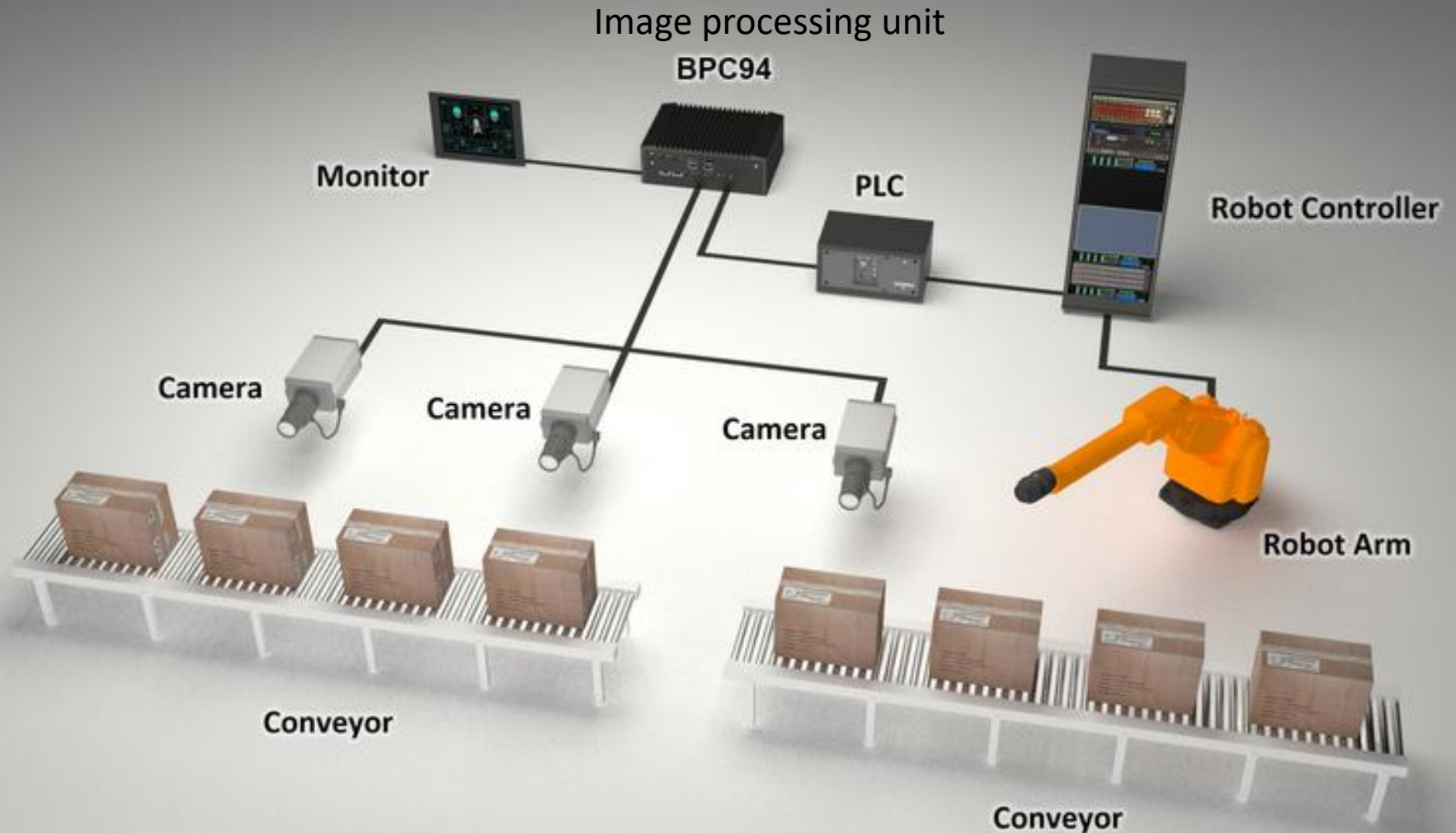
Along the complete production chain



The basic setup

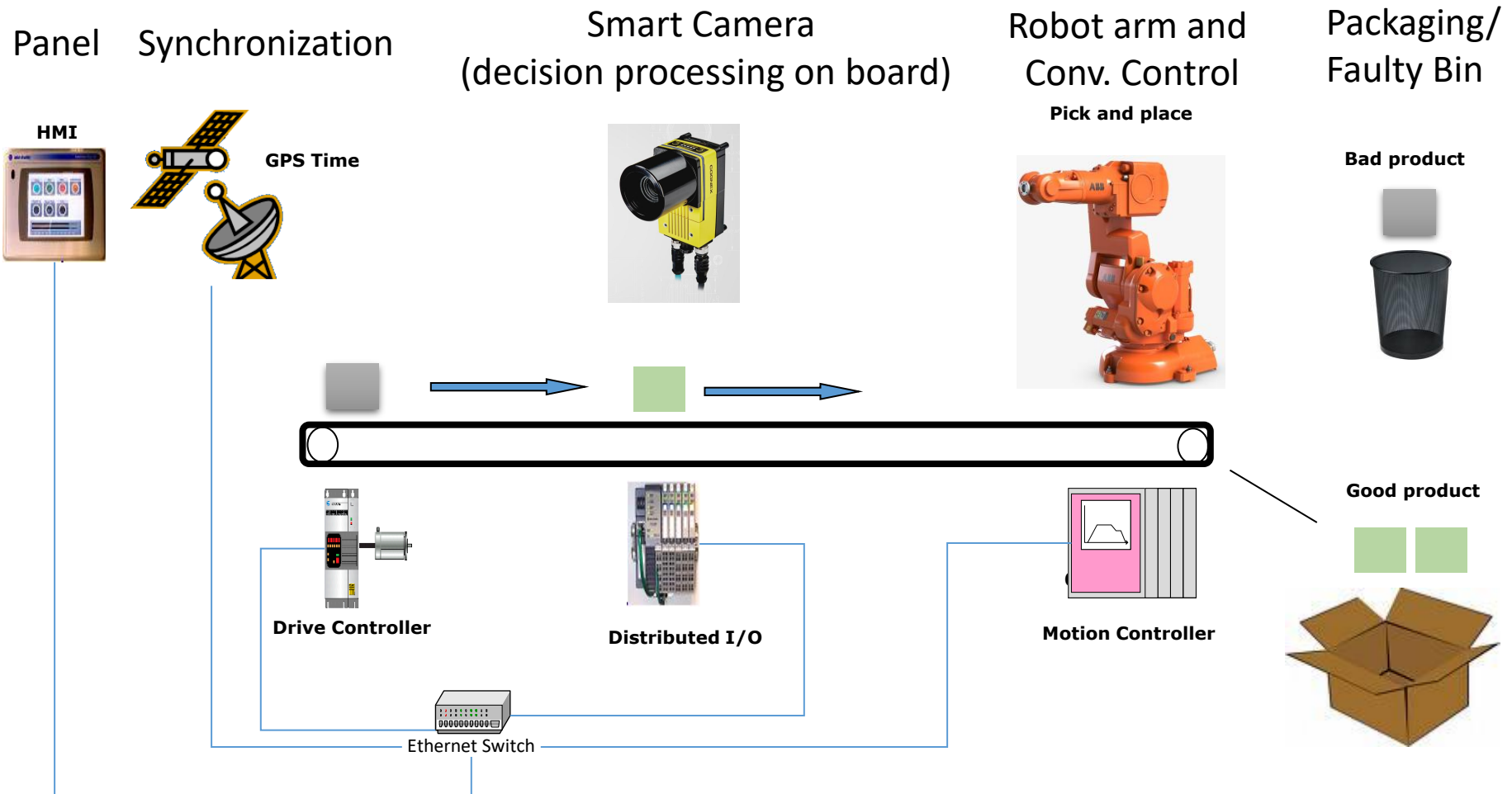


Industrial Vision Environment

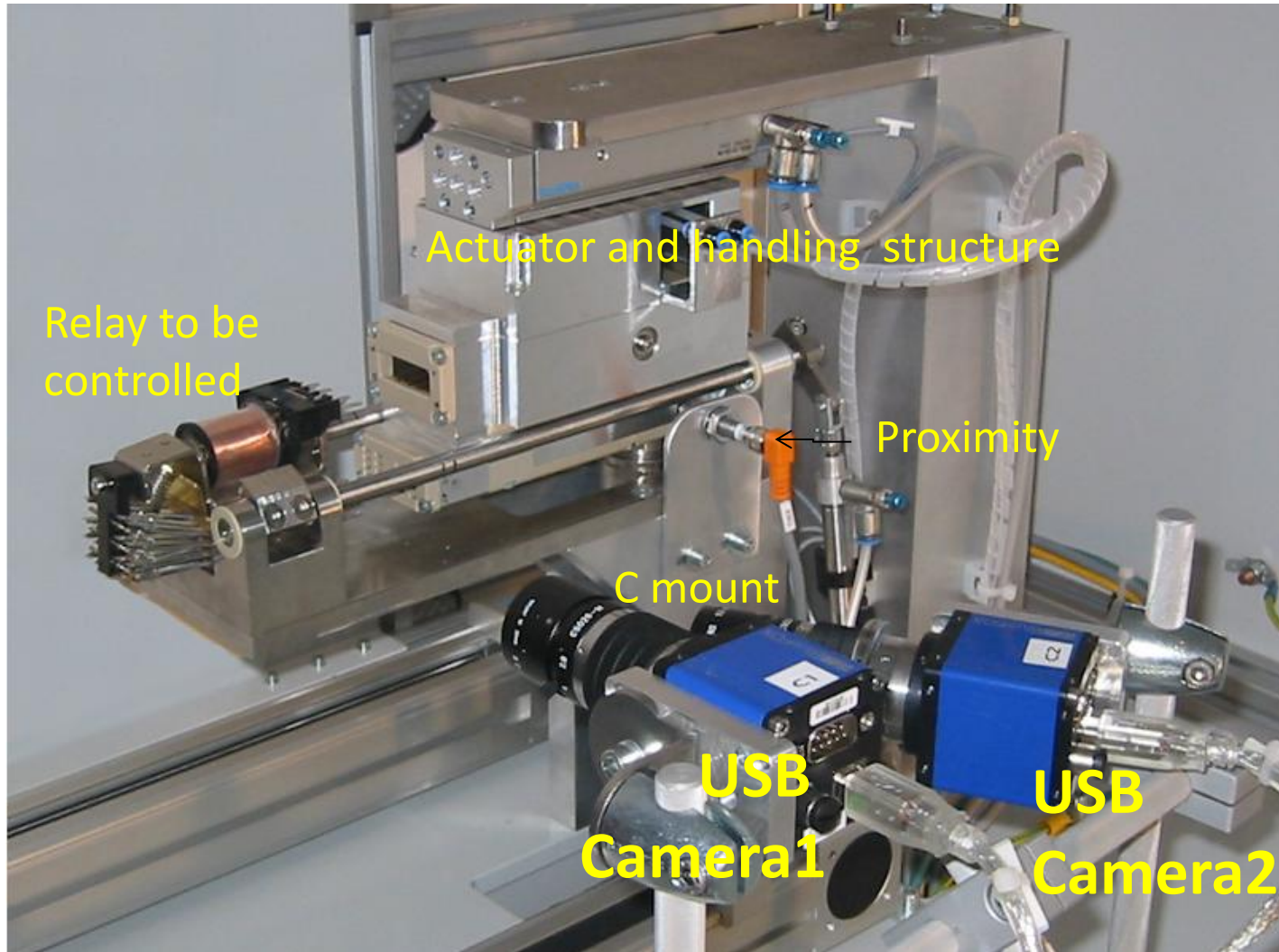


Automation Control Network Example

- An intelligent system **sorts defective parts off of a conveyor belt**
- Network uses fieldbus protocols on a 10/100 Ethernet network with IEEE 1588 time synchronization support



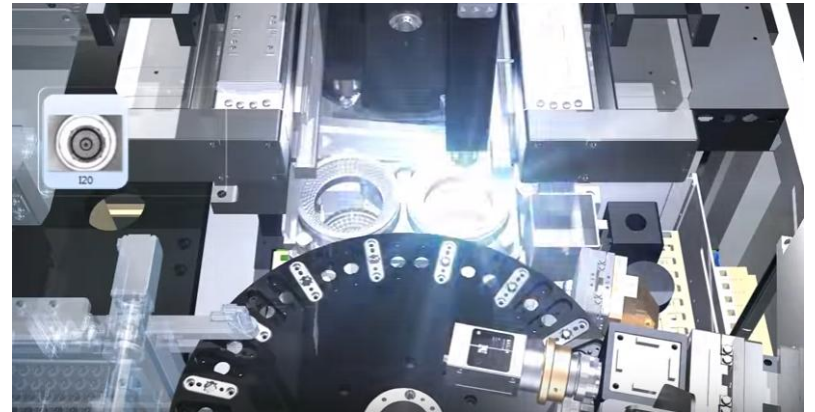
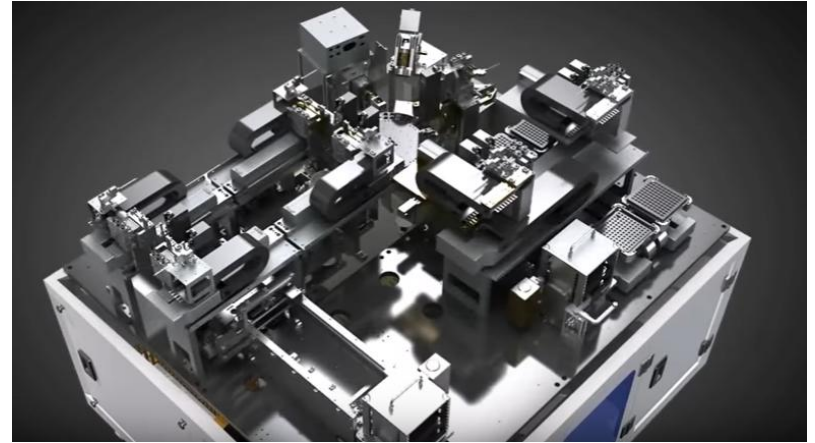
Example of industrial vision system: Relay contact measurement



Complete systems



100.000 – 200.000 EUR



Anomaly Detection

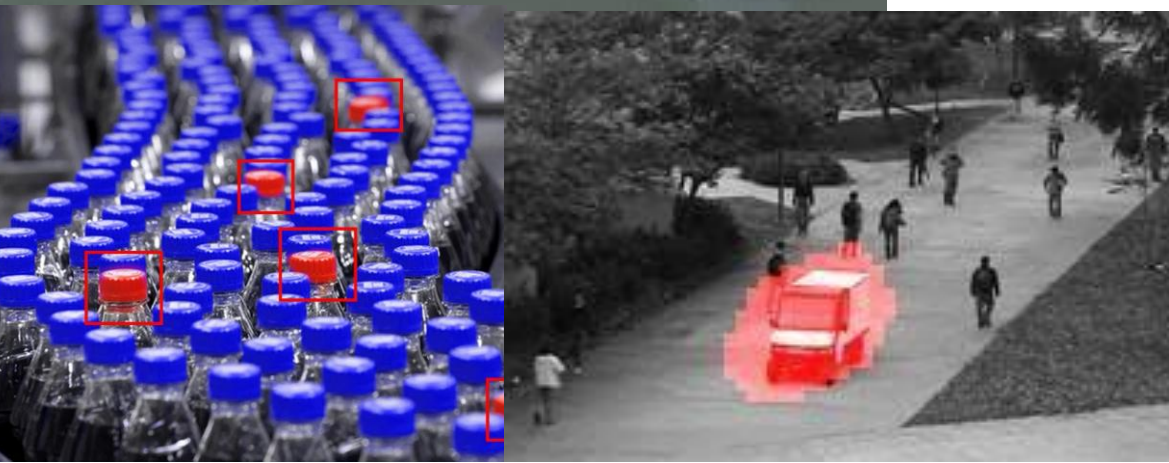
The identification of rare events, items, or observations which are suspicious because they differ significantly from standard behaviors or patterns.



How can a camera be taught to detect deviations if they are not or not completely predictable?

Rule-based image processing would have to capitulate. With the AI systems such a challenge can easily solved.

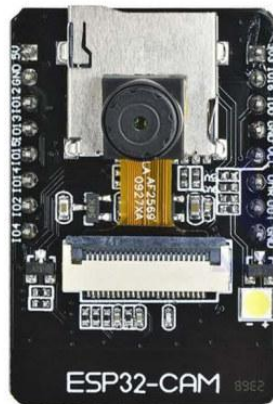
Anomaly detection is third AI functionality in addition to object detection and classification.



Small projects/budgets



EDGE IMPULSE



ESP32-CAM

CPU: Dual core, 32-bit,
240 MHz, 4 MB Flash,
520 KB SRAM
Camera: 2 MP
Price: ≈ \$ 8



Sipeed MAIX Bit

CPU: RISC-V Dual Core,
64-bit, 400 Mhz, 16
MB Flash, 8 MB SRAM
Camera: 2 MP
Price: ≈ \$ 21



M5StickV AI Camera

CPU: RISC-V, 64-bit,
400 MHz, Dual FPU, 16
MB Flash, 8 MB SRAM
Camera: VGA
Price: ≈ \$ 30



OpenMV Cam H7

CPU: ARM Cortex-M7,
32-bit, 480 MHz, 2 MB
Flash, 1 MB SRAM
Camera: 5 MP
Price: ≈ \$ 65

2MP!
8 USD

Exam simulation

- A simulation of the exam is now given in class. As explained in the first lesson, no pdf will be distributed.
- See the simulation @lesson #4 and #6 about questions regarding the first 4 and 6 lessons
- See the simulation @lesson #13 about questions regarding lessons 4-13
- In this simulation we will focus on lessons #14-24

Main points



- Applications of Intelligent Vision Systems
 - Industry 4.0 and 5.0
 - Ambient Intelligence and Domotics
 - Supply Chain
 - Environmental
- Intelligent Vision Systems
 - Object detection, Segmentation and Recognition
 - Classical, Deep and Hybrid solutions
 - Use cases
- Appendix: Industry 4.0 details (after this slide)



Industry 4.0

Appendix

(not in the exam)

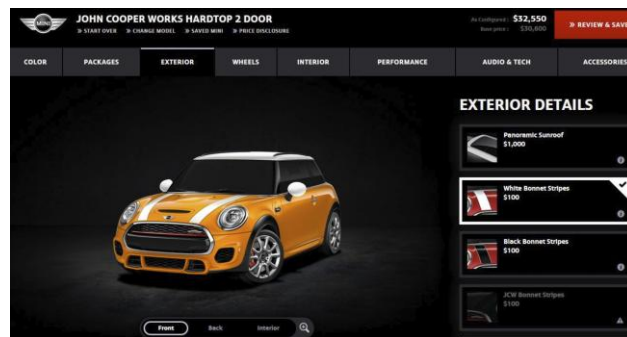
Industry 4.0

4 main synergic features



1) Vertical networking of smart production systems

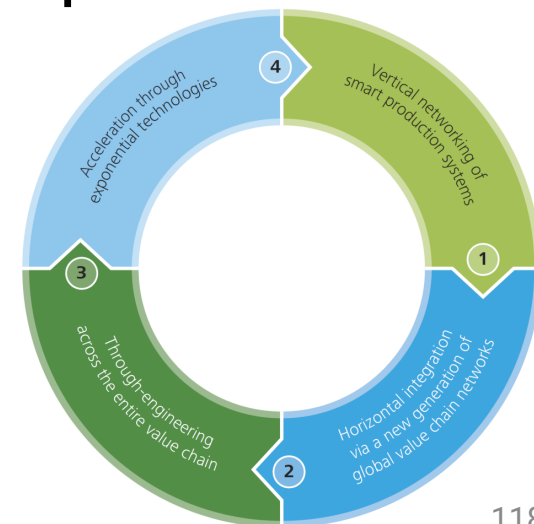
- This vertical networking uses **cyber-physical production systems (CPPSs)** to enable plants to react rapidly to changes in demand or stock levels and to faults.
- Smart factories organize themselves and enable production that is customer-specific and personalized.



Customers/market



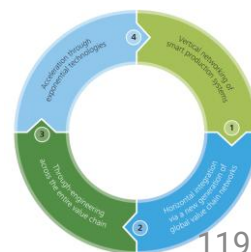
Production



2) Horizontal integration via a new generation of global value chain networks



- Local and global networks provide **networking** via CPPSs from logistics, warehousing, production, marketing and sales.
- Networks are real-time optimized
→ transparency, flexibility to respond more rapidly to problems and faults facilitating better global optimization.
- The history of any part or product is **logged** and can be accessed at any time, ensuring constant traceability.



3) Through-engineering across the entire value chain

The development and manufacture of

- new products
- and production systems

is integrated and coordinated with product life cycles, enabling new synergies to be created between product development and production systems.

development $\leftrightarrow$ manufacture $\leftrightarrow$ product life cycles



4) Acceleration through exponential technologies

- Industry 4.0 already requires automation solutions to be
 - highly **cognitive**
 - and highly **autonomous**
- To increase autonomy further still and to speed up individualisation and flexibilization:
 - Artificial intelligence
 - Advanced robotics
 - Sensor technology

